

Perovskite Photovoltaics in 2025: The certified frontier held while verification stayed scarce

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Abstract

This review assembles the complete 12-month publication record retrospectively, rather than through sequential monthly monitoring. The corpus holds 7931 publications meeting the topical scope criteria, of which 450 were examined in close detail across date-resolved monthly cohorts, alongside 1089 publications dated only to the calendar year. Within this literature, only 66 investigations reported an independently certified power conversion efficiency (PCE), which peaked at 33.15% for the year. Certified values moved from 30.26% in the first month recording a verified metric to 30.1% in the last, a net displacement of -0.16. Buried contact modification with carbazole self-assembled monolayers (SAMs), low-dimensional capping heterostructures, and additive-directed crystallization drove the performance records in both monolithic tandem and single-junction architectures. Mechanistic consensus breaks down over whether these surface modifiers permanently eliminate deep electronic traps or merely impose transient electrostatic potential shifts. Several groups attribute suppressed non-radiative recombination and stabilized open-circuit voltage (Voc) to chemical defect healing, while competing investigations demonstrate that asymmetric interfacial dipoles create electrostatic crosstalk, transport barriers, and unshielded internal fields that accelerate halide segregation. Carrying these interfacial treatments to scalable modules incurs a severe area penalty, with sharp efficiency deficits as aperture dimensions expand toward the maximum reported area of 764 cm². In blade and slot-die deposition, the molecular attributes that minimize interface recombination in spin-coated cells directly impair meniscus translation, because hydrophobic self-assembled monolayer (SAM) terminal groups form precursor micellar aggregates that induce solvent dewetting, pinhole formation, and lateral nucleation gradients across wide substrates. Operational stability data remain deficient across the publication record. Merely 15 studies documented operational lifetimes, the longest reaching 3200 h, and only 24 articles followed standardized International Summit on Organic Photovoltaic Stability (ISOS) procedures. Most stability assessments evaluated unencapsulated storage in inert atmospheres, or omitted continuous maximum power point tracking (MPPT) under elevated operating temperatures, which leaves lifetime projections uncalibrated against operational degradation modes. Abstract-level reporting frequencies confirm the same verification gap: statements of independent certification occurred in just 5.8% of evaluated abstracts across the year, varying between

3.4% and 7.6% across monthly cohorts. Champion efficiency claims therefore rest predominantly on unverified, small-area laboratory cells that remain susceptible to spectral mismatch and calibration artifacts. Read across device architectures, the high open-circuit voltages obtained through contact engineering are frequently purchased with accelerated chemical and electronic degradation. Phosphonic acid anchor groups corrode reactive nickel oxide hole-transport layers under operating bias, and low-dimensional perovskite capping overlayers introduce anisotropic out-of-plane transport barriers that accumulate photogenerated carriers, lowering the kinetic activation barrier for electric-field-driven halide migration and phase demixing. In narrow-bandgap tin-lead absorbers, evidence settling whether stannous oxidation stems from autonomous chemical redox reactions or from downstream lattice microstrain remains thin, since chemical reducing agents delay degradation without resolving the underlying mechanical distortion. Establishing whether passivating molecules permanently eliminate deep recombination centers, or merely raise temporary barriers that disguise latent operational instabilities, requires synchronized operando spectroscopic and impedance measurements under simultaneous electrical bias and thermal stress.

Keywords: perovskite photovoltaics; single-junction cells; all-perovskite tandems; perovskite/silicon tandems; modules; operational stability; annual review

1. Introduction

Perovskite photovoltaic research in 2025 exposed stark disagreements where independent investigations reached incompatible conclusions regarding identical physical degradation pathways. While several studies identified mobile iodide accumulation at electrode interfaces as the primary driver of rapid delamination, others attributed initial performance losses to irreversible redox reactions with adjacent charge-transport layers that proceed independently of halide migration. These competing models remain unresolved because conventional ex situ characterization decouples applied electrical bias from local chemical transformation. Settling this conflict requires operando, depth-resolved chemical spectroscopy synchronized with local potential and impedance mapping under simultaneous operational bias and thermal stress. This divergence illustrates how the field’s central challenge has shifted: progress is no longer defined by incremental gains in power conversion efficiency (PCE), but by the interdependent demands of verification, scalable area, and operating lifetime.

Across the published literature, 7931 indexed works met the scope gate and 450 were read closely. This review was compiled retrospectively from the complete 2025 publication record, not by monitoring the literature month by month during 2025. Within this intake, 12 months carry date-resolved works and 1089 works are dated only to the year, so they contribute to the corpus but not to any month-by-month series. Because this collection forms an initial edition without a prior yearly issue, no year-over-year comparisons are made.

The mechanistic evaluation follows four primary threads. The first addresses single-junction devices to identify where the remaining losses sit, establishing whether non-radiative recombination concentrates at buried substrate contacts, bulk defect states, or top extraction interfaces. The second examines all-perovskite tandems, evaluating tin-lead (Sn-Pb) stability and subcell matching to resolve how divalent tin oxidation competes with phase segregation under current-matched illumination. The third considers hybrid tandems by assessing the interfacial coupling of perovskite absorbers to silicon, copper indium gallium selenide (CIGS), and organic subcells, balancing optical management against processing damage and mechanical strain. The fourth examines modules and large-area devices through the fluid dynamics and crystallization kinetics of scaling the deposited film, where

drying rates and solvent extraction govern defect distributions across large substrates.

Throughout each thread, this review distinguishes what the year’s literature verified from what it asserted. Reported PCE values measured on sub-0.1 cm² apertures without independent verification, lifetime projections derived without continuous maximum power point tracking under load, and passivating mechanisms inferred without operando confirmation are separated from claims supported by reproducible physical measurements.

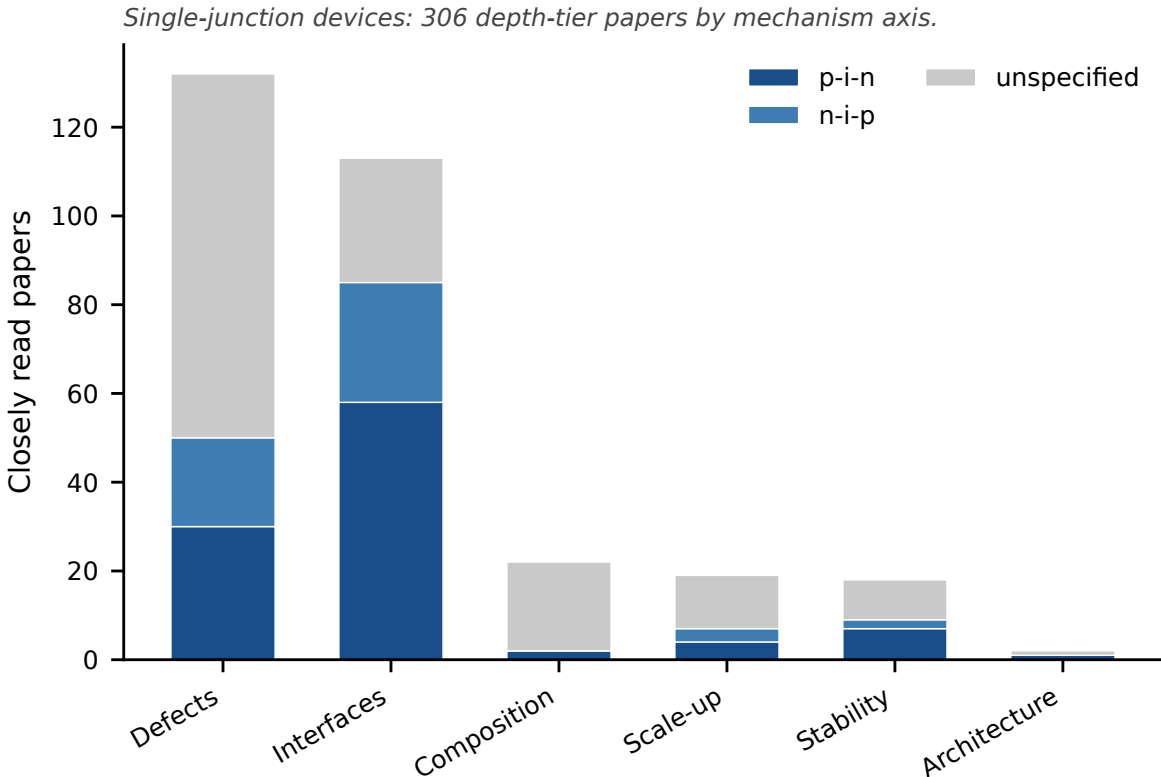


Figure 1. Single-junction devices: where the year’s close reading was directed, by mechanism axis and device architecture.

2. The Certified Frontier Across 2025

Conflicting investigations across the year dispute how surface molecular passivation and low-dimensional heterojunctions behave at wide-bandgap (WBG) perovskite interfaces. One camp holds that they eliminate non-radiative recombination without transport penalties, while the other holds that they induce severe electrostatic crosstalk. Capping WBG absorbers with three-dimensional/quasi-two-dimensional perovskite heterojunctions suppresses halide segregation and passivates interface defects, which minimizes open-circuit voltage (Voc) deficits. That stack delivers an independently certified power conversion efficiency (PCE) of 28.81% in two-terminal (2T) tandems [1], and a related device reached 28.52% certified in an unknown device class [2]. Modulating competitive adsorption of hybrid self-assembled monolayer (SAM) molecules at buried contacts aligns interfacial energy levels, achieving a certified 2T tandem PCE of 28.78% [3]. Directional nucleation via ammonium additives yields 28.06% in 2T tandems [4]. In sharp disagreement, competing reports demonstrate that asymmetric cation surface treatments generate

electrostatic potential distortions and chemical crosstalk that impede carrier extraction across hole-selective contacts. Those reports conclude that only self-consistent cation-anion integration eliminates this crosstalk, reaching a certified 2T tandem PCE of 31.47% [5]. Whether surface modifiers permanently eliminate deep traps or merely establish transient electrostatic screening remains open, and settling it requires *operando*, bias-dependent drive-level capacitance profiling (DLCP) paired with intensity-modulated photovoltage spectroscopy across functional junctions under operating electrical load [1,5].

A parallel divergence divides investigations of narrow-bandgap mixed tin-lead (Sn-Pb) subcells. At issue is whether operational degradation is governed primarily by mobile halide defect migration or by mechanical microstrain. Chemical immobilization strategies treat vacancy-assisted ion drift as the primary failure mode, and on that view coordinating mobile halides and metal centers suppresses defect generation, achieving certified 2T tandem PCE values of 28.31% [6] and 27.17% [7]. Competing studies demonstrate instead that chemical passivation alone cannot prevent degradation, because lattice mismatch and localized microstrain accelerate phase decomposition. Introducing lattice-stabilizing networks to homogenize internal strain and decouple photothermal-mechanical breakdown yields a certified PCE of 28.9% in 2T tandems [8]. Related networks reach 28.11% in 2T tandems [9] and 29.2% in an unknown device class [10]. Differentiating chemical defect migration from lattice strain relaxation as the rate-limiting degradation pathway remains open, and doing so requires *operando* high-resolution synchrotron X-ray diffraction (XRD) topography correlated with depth-resolved time-of-flight secondary ion mass spectrometry (ToF-SIMS) under simultaneous electrical bias and thermal stress [6,8,10].

Conformal growth on textured substrates reveals a third divergence between physical substrate modification and chemical crystallization control. One route abandons standard commercial textures: re-engineering industrially textured silicon into iceberg-like rounded pyramids enables uniform solution wetting and conformal absorber deposition, establishing the annual certified record of 33.15% in 2T tandems [11]. Alternative approaches show that altering substrate topography is unnecessary if crystallization kinetics and nucleation dynamics are modulated chemically. Oriented grains on unflattened textures achieve certified 2T tandem efficiencies of 31.4% [12], 30.26% [13], and 27.92% [14]. Monolithic integration with copper indium gallium selenide (CIGS) bottom subcells introduces inter-subcell mechanical coupling. Here substrate corrugation relieves tensile stress to achieve a certified 2T tandem PCE of 28% [15]. Composite intermediate recombination layers achieve certified 2T tandem efficiencies of 30.1% [16] and 25.45% [17]. However, macroscale area expansion drops certified 2T tandem efficiency to 28.9% [18]. Certified modules fall further, to 25.2% [19] and 22.06% [20]. That spread exposes the persistent gap between small-area laboratory cells and scalable tandems [11,18].

Between the first month with data (30.26% in 2025-01 for 2T tandems) [13] and the final month (30.1% in 2025-12 for 2T tandems) [16], the net change in best certified efficiency was -0.16 percentage points. The monthly series displays neither steady progress nor monotonic stagnation, showing instead punctuated dispersion across 66 certified reports out of 450 closely read papers. Certified monthly maximums dropped to 27.17% in 2025-02 (4 certified papers, 2T tandem) [7], and they rebounded to 28.78% in 2025-03 (6 certified papers, 2T tandem) [3] and 28.31% in 2025-04 (4 certified papers, 2T tandem) [6]. Certified reporting then ceased in 2025-05 (0 certified papers), before reaching 28.81% in 2025-06 (3 certified papers, 2T tandem) [1]. After dropping to 26.56% in 2025-07 (8 certified papers, inverted p-type-intrinsic-n-type (p-i-n) single-junction) [21], certified performance

peaked at 33.15% in 2025-08 (6 certified papers, 2T tandem) [11]. It then retreated through 31.47% in 2025-09 (7 certified papers, 2T tandem) [5], 28.9% in 2025-10 (7 certified papers, 2T tandem) [8], and 26.96% in 2025-11 (7 certified papers, p-i-n single-junction) [22], closing at 30.1% in 2025-12 (5 certified papers, 2T tandem) [16]. Concurrently, certified single-junction p-i-n devices converged between 26.79% [23], 26.74% [24], and 26.65% [25]. Uncertified self-reported claims outpaced certified metrics in eleven of twelve months, and they ran up to 33.2% in 2025-05 and 33.5% in 2025-12 (both in unclassified device architectures), coinciding with certified data only in 2025-08 at 33.15%. Crucially, multiple publications carry only the year 2025 without month designations. Those papers are excluded from the series, which therefore does not represent complete coverage of annual output.

Table 1. Best independently certified and best self-reported power conversion efficiency in each device class, with the detailed-balance limit that applies to that class. Certified and self-reported values are kept in separate columns because they are not the same kind of evidence. Simulation and review studies contribute no value to this table.

Device class	Best certified (%)	Best self-reported (%)	Detailed-balance limit (%)	Papers with a certified value	Papers read
Single-junction cells	26.96	29.11	29.4	29	332
All-perovskite tandems	29.2	30.69	47.6	8	28
Perovskite/silicon and other hybrid tandems	33.15	33.5	47.6	12	47
Modules and large-area devices	28.78	28.94	47.6	16	43

3. Single-Junction Devices: Where the Remaining Losses Sit

Single-junction perovskite solar cells (PSCs) with a 1.55 electron-volt (eV) absorber bandgap face a theoretical Shockley-Queisser (SQ) limit of 29.4%, yet certified records stand at 26.96% power conversion efficiency (PCE) with an uncertified champion of 27.02% and stabilized output of 26.61% on a 1 cm² aperture under continuous 1-sun maximum power point (MPP) tracking [22]. Across 332 investigations evaluated this year in this device class, 278 reporting measured metrics, the field divides sharply over where the remaining ~2.4% absolute efficiency deficit resides. One body of evidence attributes open-circuit voltage (VOC) and fill factor (FF) deficits to the buried hole-selective contact, citing energetic misalignment, pinholes, and chemical degradation of carbazole-based self-assembled monolayers (SAMs). An opposing body contends contact extraction is optimized and that losses reside in the absorber bulk and at grain boundaries through point defects, uncoordinated lead ions (Pb²⁺), and residual strain. This divergence drives conflicting engineering: one camp modifies substrates while another leaves contacts untouched, achieving identical gains via bulk crystallization. Settling whether primary losses sit at the contact or in the bulk requires operando absolute-intensity photoluminescence (PL) spectroscopy and cross-sectional quasi-Fermi level splitting (QFLS) mapping under 1-sun operational bias. Correlating internal QFLS with external terminal voltage across operating cells, rather than uncontacted precursor films, is the only measurement that isolates interfacial recombination velocities from bulk non-radiative decay.

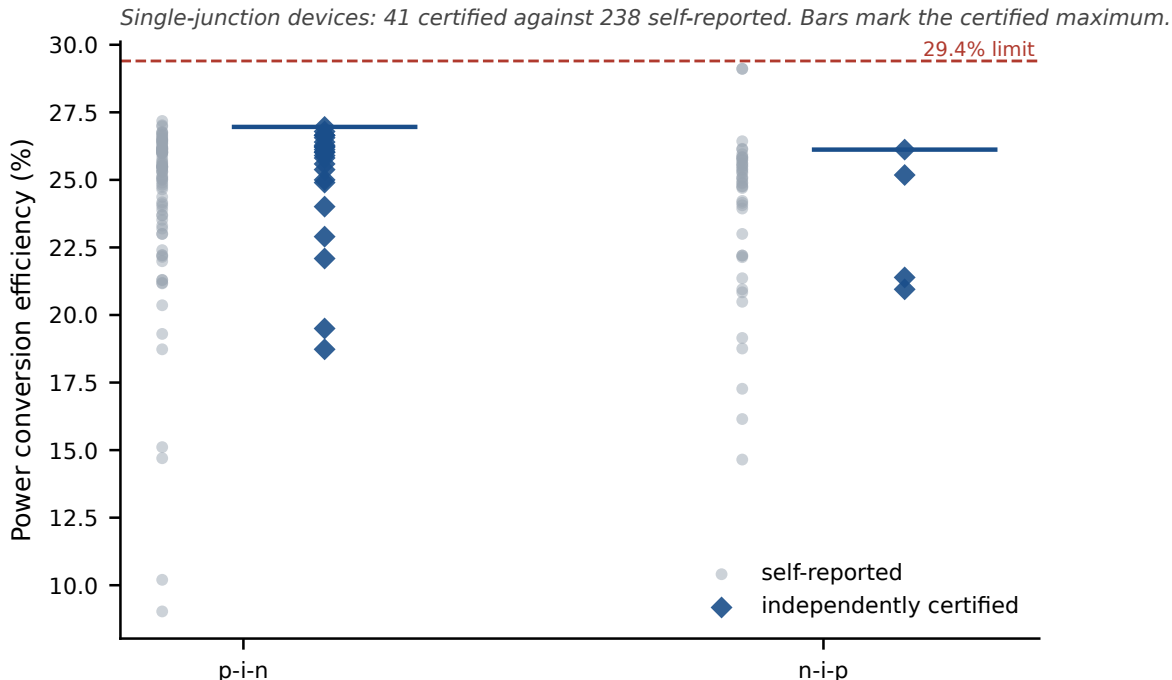


Figure 2. Single-junction devices: independently certified against self-reported champion efficiencies, grouped by device architecture. The dashed line marks the 29.4 percent Shockley-Queisser limit for a 1.55 eV absorber. Simulation and review studies are excluded.

The dispute over the buried hole transport layer (hole-transport layer (HTL)) contact exposes structural contradictions in molecular design. In inverted (p-i-n) architectures with SAMs on indium tin oxide (ITO) or nickel oxide (NiO_x), one group asserts molecular rigidity is mandatory to enforce vertical alignment, packing density, and order. The rigid pyrene self-assembled monolayer (SAM) PhPAPy adopts vertical orientation on ITO, yielding a certified reverse-scanning PCE of 26.74% and certified stabilized output of 26.12% under International Summit on Organic Photovoltaic Stability (ISOS)-L-2 protocol [24]. Rigid conjugated bisphosphonic acids such as TPA2P exhibit a deep highest occupied molecular orbital (HOMO) of -5.47 eV, strengthening ITO binding and suppressing pinholes to yield an uncertified 26.11% PCE [26], while all-conjugated block copolymers providing rigid transport channels reach an uncertified 26.11% PCE [27]. Conversely, another set of investigations concludes rigid backbones induce steric embrittlement and pinhole dewetting, arguing flexible spacers are needed to relieve interfacial stress. Combining flexible aliphatic heads with rigid aromatic linkers in (4-(diphenylamino)phenyl)phosphonic acid yielded an uncertified champion PCE of 26.21% on 0.0715 cm^2 , but collapsed to 24.49% on 1 cm^2 [28]. Likewise, a flexible central substituent in pyridine derivative 2PYET achieved an uncertified 26.03% PCE on 0.09 cm^2 , but dropped to 24.45% on 1.0 cm^2 [29]. These severe efficiency drops upon scaling from sub- 0.1 cm^2 to 1 cm^2 show that micro-cell records often mask spatial dewetting that rigid versus flexible designs fail to reconcile.

The chemical stability of the SAM-substrate junction presents a parallel loss pathway. On NiO_x , redox reactions between electrophilic Ni^{3+} sites and phosphonic acids trigger de-anchoring and interfacial traps. Countering this by blending MeOF-4SHCz with MeOF-4PACz (4:1 weight ratio) forms sulfur-oxygen-nickel (S-O-Ni) bonds at Ni^{3+} -rich regions, achieving a certified PCE of 26.28% (uncertified champion 26.5%) with $T_{90} > 1000$ hours [30]. Covalent tridentate anchoring

via (3,3,3-trifluoropropyl)triethoxysilane (3F-PTES) on NiO_x eliminates hydroxyl defects and produces an energetic offset of 0.01 eV with the perovskite, achieving an uncertified 26.47% PCE [31]. Pre-anchoring arginine on NiO_x weakens SAM amphiphilicity, suppressing aggregation to yield an uncertified 26.67% PCE on a 0.049 cm^2 aperture [32]. Multi-component assembly targets aggregation similarly: co-assembling benzylphosphonic acid (BPPA) with Me-4PACz coordinates NiO_x defects and chelates interfacial Pb^{2+} (uncertified champion 26.35%) [33], while embedding covalent organic frameworks (COFs) PT-COF and Zn-PT-COF into MeO-4PACz disrupts aggregation via pi-pi and hydrogen bonding, achieving a certified 26.12% PCE (uncertified champion 26.39%) [34]. Aggregation is also suppressed by molecular doping with MCC, delivering a certified 26.65% PCE (champion 26.78%) with stability exceeding 2000 h and bifacial module efficiencies of 15.51% and 14.64% [35], functionalized ITO nanoparticles with -OH groups resisting solvent rinsing under ISOS-L-2 testing (uncertified champion 26.44%) [36], and co-assembling acetyl dl-carnitine chloride with Me-4PACz (uncertified champion 25.78%) [37]. Photochemical degradation of phosphonic SAMs under ultraviolet light prompted polymeric networks such as Poly-2PACz, yielding an uncertified champion PCE of 26.0% with a T80 of 500 h under ultraviolet illumination [38]. Interfacial dipole engineering with 2DPAA enlarges the buried dipole moment to 5.10 D, securing a certified 26.02% PCE (uncertified champion 26.53%) [39], while electrostatically enhanced SAM anchoring achieves a certified 26.23% (steady-state 25.42%) on 1 cm^2 and 24.72% on a 20.7 cm^2 module [40].

Molecular interlayers and chelating bridges inserted between the SAM and perovskite absorber attempt to bridge interfacial mismatch. An aluminum glycinate interlayer coordinates unanchored P-OH and NiO-OH while chelating Pb^{2+} , securing a certified 26.21% PCE (uncertified champion 26.74%) in 1.55 eV cells and 20.71% in 1.785 eV wide-bandgap cells [41]. Organic bridge BFS aligns energy levels and chelates Pb^{2+} , raising uncertified efficiency to 26.16% from a 24.49% control [42], while 1,3,5-triazine-2,4,6-tricarboxylic acid derivative (TDCA) bridges achieve an uncertified 26.15% PCE (26.20% for 1.55 eV and 21.65% for 1.77 eV absorbers) [43]. Incorporating 1-benzyl-3-methylimidazolium tetrafluoroborate (BzMIMBF₄) establishes pi-conjugated heterojunctions passivating interface traps (uncertified champion 26.45%) [44], while 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ) forms charge-transfer complexes that raise VOC to 1.202 V with an uncertified 25.91% PCE and 91% retention after 1000 h maximum power point tracking (MPPT) [45]. Inserting 1,1-bis(4-di-p-tolylaminophenyl)cyclohexane (TAPC) increases FF from 83.2% to 85.2%, reaching an uncertified 26.05% PCE on rigid substrates and 24.39% on flexible substrates [46], while phenylhydrazinium chloride passivates defects and retards crystallization to reach an uncertified 25.80% PCE from a 24.67% control [47]. In regular (n-i-p) devices, buried interface losses at tin oxide (SnO_2) electron transport layer (electron-transport layer (ETL)) substrates involve oxygen vacancies and poor wetting. L-citrulline molecular bridges achieve an uncertified 25.95% PCE on a 0.07065 cm^2 aperture [48]. Potassium trifluoro(sulfonatomethyl)borate (TFSK) passivates SnO_2 oxygen vacancies and hydroxyls, reaching an uncertified 25.82% PCE [49], while dipole-functionalized molecules targeting dual-site buried defects yield an uncertified 25.85% PCE (24.79% on 1 cm^2) [50]. Whether these interlayers primarily suppress interface trap recombination or template superior crystal growth remains unresolved due to absent depth-resolved recombination lifetimes.

This uncertainty connects directly to absorber bulk losses and the contentious role of two-dimensional (2D) perovskite phases at grain boundaries. Two conflicting schools debate whether converting residual lead iodide (PbI_2) into 2D phases benefits performance. The prevailing view treats PbI_2 as a defect source requiring conversion into 2D phases to suppress non-radiative decay and block

mobile ions. Guanfacine hydrochloride conversion of boundary PbI_2 into 2D perovskite achieves an uncertified 26.19% PCE [51]. 1-(pyren-1-yl)methanamine hydroiodide (PyPAI) semiconductor spacers induce columnar 2D stacks directing vertical crystallization, reaching an uncertified 26.41% PCE [52]. Tilted-oriented 2D perovskites enhance conductivity and uniformity, securing a certified 26.3% PCE (uncertified champion 26.6%) with 90% retention after 1100 h MPPT [53], while sequential ligand treatment using 2,6-difluorobenzylammonium bromide (DFABr) at boundaries and 4-(aminomethyl)piperidinium bromide (4APPBr2) on surfaces yields an uncertified 26.06% PCE [54]. For indoor photovoltaics, Dion-Jacobson/Ruddlesden-Popper (DJ/RP) dual 2D heterostructures using butylammonium iodide (BAI) achieve an uncertified 30.3% PCE with a twentyfold T90 lifetime gain [55]. In direct conflict, an opposing study proves 2D perovskite phases introduce insulating out-of-plane transport barriers that impede extraction and promote charge accumulation. Introducing carboxymethyl-beta-cyclodextrin (CM-beta-CD) to suppress phenylethylammonium iodide (PEAI) deprotonation deliberately inhibited 2D phase formation, raising PCE to an uncertified 26.06% from 23.75% for control devices, with 91.2% retention after 1050 h at 25 °C and 25-35% relative humidity (RH) [56]. This contradiction shows that 2D phase engineering often trades carrier transport for chemical encapsulation. Settling this trade-off requires temperature-dependent transient photovoltaic and drive-level capacitance measurements to quantify extraction barriers across boundaries.

Where 2D phase conversion is avoided, eliminating bulk non-radiative decay relies on additives that regulate crystallization, chelate defects, and relax strain. Dimethyl sulfide soft Lewis bases guide heterojunction formation via soft-soft interactions, securing a certified 26.48% PCE (uncertified champion 26.7%) and over 94% retention after 2000 h MPP tracking under ISOS-L-1 [57]. Coordinating Pb^{2+} and hydrogen bonding with iodide (I-) and formamidinium (FA+) using sodium 2-amino-4-(methylsulfonyl)butanoate (NHMS) achieves a certified 26.35% PCE (uncertified champion 26.51%) [58], while dual-ligand regulation with PCH and PBD enforces homogeneous ion distributions to deliver a certified 26.33% PCE (uncertified champion 26.66%) [59]. Inhibiting light-induced decomposition with a hindered amine light stabilizer (HASS) yields a certified 26.56% PCE (uncertified champion 26.74%) [21]. Dual-site passivation with AGAD and AESM via vacuum flash evaporation eliminates boundary defects, reaching an uncertified 26.14% PCE with 90% retention after 1200 h MPPT [60]. Suppressing the non-perovskite 6H hexagonal intermediate with butylamine acetate (BAAc) promotes alpha-phase FAPbI_3 crystallization, reaching an uncertified 26.12% PCE [61]. Introducing phthalocyanine derivative BE-ZnPc coordinates Pb^{2+} , yielding a certified 26.05% PCE (uncertified champion 26%) with 92% retention after 550 h at MPP [62], while diammonium ligands CyDMADI and PhDMADI exploit electron-withdrawing cores to passivate bulk traps, reaching an uncertified 26.04% PCE [63]. Relieving lattice strain with 3-methylthio-1,2,4-triazine yields an uncertified 26.04% PCE (24% on flexible substrates) [64], whereas 4,4'-oxybis(benzenesulfonyl chloride) (OBSC) binds Pb^{2+} through Pb-O interactions to yield a certified 26.03% PCE (uncertified champion 26.39%) [65]. Modulating FA+ and methylammonium (MA+) hydrogen bonding with multisite amide derivative NUNB achieves a certified 26.03% PCE (uncertified champion 26.34%, 21.92% on rigid modules) [66]. Dual organic potassium salts omnidirectionally passivate defects, retaining 88.7% PCE after 1100 h at 85 °C with an uncertified 25.78% PCE [67]. Decoupling lattice strain via in situ polymerization of poly(2-acrylamido-2-methylpropane sulfonic acid) (P-AMPS) provides grain spatial isolation, delivering an uncertified 25.78% PCE under ISOS-L-1 [68]. Single-crystal PSCs produced via nucleation engineering achieve an uncertified benchmark PCE of 25.8% [69]. The fact that single crystals lacking grain boundaries do not exceed polycrystalline thin-film records demonstrates that grain boundary area alone does not limit performance; point defects and contact extraction

dominate.

The third critical loss zone sits at the top electron-selective contact in inverted devices, where fullerenes (buckminsterfullerene (C_{60}) and phenyl-C61-butyric acid methyl ester (PCBM)) interface with perovskite and metal cathode layers. Under operational stress, C_{60} films suffer dimerization, pinholes, and low fracture toughness that permits halide ions to corrode the metal contact. Interfacial sodium-heptafluorobutyrate functionalization promotes a compact C_{60} layer blocking ion diffusion, delivering the certified record PCE of 26.96% (uncertified champion 27.02%, stabilized 26.61% on 1 cm^2) [22]. Supramolecular host-guest encapsulation with C8A suppresses ion migration to reach an uncertified champion PCE of 27.18% (certified 26.79%) in inverted cells, unsealed devices retaining 95% initial PCE after 1015 h at MPP, alongside 26.01% (certified 25.68%) for TiO_2 -based regular n-i-p devices [23]. To overcome C_{60} mechanical fragility, ionic salt CPMAC increases interfacial toughness threefold, enabling an uncertified 26% PCE on 6 cm^2 cells with $\sim 2\%$ degradation after 2100 h of 1-sun operation at 65°C [70]. In PCBM contacts, phosphate-substituted fullerene FuPE compacts stacking and boosts electron mobility to $0.183\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$, securing an uncertified 26% PCE [71]. At the perovskite/ETL junction, triphenylphosphine oxide (TPPO) coordinates Pb^{2+} , yielding an uncertified 26.01% PCE with a VOC of 1.23 V and a minimal voltage deficit of 0.32 V for MA-free PSCs under continuous MPPT [72]. Cathode modification with dipolar Rh-Py establishes ohmic contact and coordinates Pb^{2+} to yield an uncertified 25.8% PCE [73].

In regular n-i-p cells, where 2,2',7,7'-tetrakis(N,N-di-p-methoxyphenylamino)-9,9'-spirobifluorene (Spiro-OMeTAD) serves as the top HTL, losses stem from dopant migration and interface traps. Planar bipyridine-thiosumanene isomer 5BT exposes Lewis base sites accelerating hole extraction, securing a certified 26.12% PCE (uncertified champion 26.15%) with 94.0% retention under ISOS-L-2 [74]. Electron-withdrawing pyridinecarboxaldehyde oxime (PO) ligands increase perovskite ionization potential to inhibit ion generation, reaching an uncertified 26.13% PCE [75], while hole interface molecule MeS-TPA-Cbz-HAI coordinates interfacial defects with Spiro-OMeTAD to achieve an uncertified 25.83% PCE [76]. HTL-free carbon-based perovskite solar cells (HTL-free C-PSCs) using lithium carbonate (Li_2CO_3) dual passivation achieve a VOC of 1.142 V and an uncertified champion efficiency of 33.2% [77], although time-resolved confocal PL microscopy confirms unpassivated boundary traps at the perovskite/carbon interface cause severe non-radiative recombination, and this 33.2% efficiency reflects non-standard indoor or concentrated conditions. Numerical modeling of quasi-interdigitated back-contact (QIBC) PSCs predicts 26.67% PCE across 30 contact pairs [78], SCAPS-1D simulations of CsPbI_2Br with CuI predict 23.66% PCE (champion 26.13% simulated) [79], and simulations of oxide double-perovskite $\text{Ba}_2\text{AsNbO}_6$ predict 29.11% PCE [80], illustrating that idealized contacts can approach the SQ limit. However, physical devices remain throttled by the coupled interplay of 2D transport barriers, SAM de-anchoring, and fullerene mechanical degradation under operating stress.

4. All-Perovskite Tandems: Sn-Pb Stability and Subcell Matching

Investigations across the past year reveal a fundamental mechanistic disagreement over the primary degradation vector in low-bandgap tin-lead (Sn-Pb) perovskites. At issue is whether oxidation of Sn(II) to Sn(IV) is initiated by autonomous chemical redox reactions, or whether it is secondary to photothermal-mechanical lattice strain that generates defect pathways. One school of thought treats Sn(II) oxidation as an intrinsic chemical defect process driven by volatile halogen species and residual impurities. That view requires active reducing agents or self-healing redox shuttles to restore

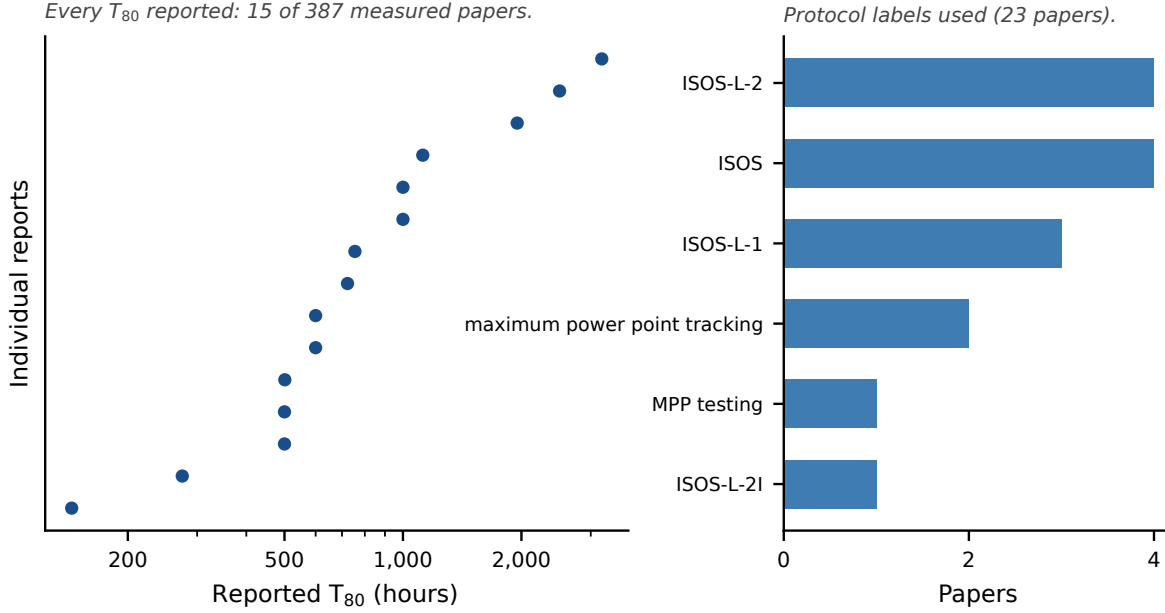


Figure 3. The year’s operational-stability evidence in full. Left, every reported time to 80 percent of initial performance. Right, the stability protocol labels used.

metal states [6,81]. Tris(2-carboxyethyl)phosphine (TCEP) exemplifies this approach, operating as an in situ redox shuttle that coordinates metal cations to enable self-healing. The shuttle allows single-junction Sn-Pb perovskite solar cells (PSCs) to reach a power conversion efficiency (PCE) of 23.86% and retain 90% of initial performance after 563 h of maximum power point tracking (MPPT), translating into an uncertified two-terminal (2T) tandem PCE of 28.86% [81]. Similarly, hydrazine monohydrochloride bulk passivation immobilizes ions via hydrazinium ($N_2H_5^+$) hydrogen bonding to iodide, while chloride fills iodide vacancies. That route achieves a 23.21% low-bandgap single-junction PCE, a certified 2T tandem efficiency of 28.31%, and an uncertified champion of 28.55% [6]. Conversely, competing investigations demonstrate that Sn(II) oxidation is predominantly downstream of photothermal-mechanical strain, where local lattice distortions generate the vacancies that subsequently facilitate oxidation [8,9,10]. Within this mechanical framework, introducing sodium 1,3,6-naphthalenetrisulfonate (NTS) forms Sn-trisulfonate coordination that strengthens Sn-I bonds and homogenizes strain. Reducing light-induced lattice distortions yields a single-junction Sn-Pb PCE of 23.2%, a certified 2T tandem PCE of 29.2%, and an uncertified champion of 29.6% [10]. The polyamine ligand tris(4-aminophenyl)amine (TPTA) establishes an I-Sn-N coordination framework that decouples photothermal-mechanical degradation from chemical breakdown. The resulting cell reaches a 23.4% single-junction Sn-Pb PCE that retains 94.9% of initial performance after 950 h of maximum power point (MPP) tracking, with a certified 2T tandem efficiency of 28.9% and a champion of 29.6% [8]. In methylammonium (MA)-free absorbers, rubidium acetate suppresses Sn(II) oxidation by alleviating internal microstrain rather than relying on sacrificial redox chemistry. That approach delivers a single-junction Sn-Pb PCE of 23.02% with an open-circuit voltage (VOC) of 0.897 V and a fill factor (FF) of 80.64%, which yields a certified 2T tandem PCE of 28.11% (champion 29.33%) [9]. Whether lattice distortion precedes vacancy generation or chemical oxidation initiates structural relaxation is still unsettled, and settling the question requires operando depth-resolved synchrotron X-ray diffraction combined with transient chemical spectroscopy under simultaneous

thermal, electrical, and optical stress. This measurement would reveal whether strain relief halts Sn(IV) accumulation in the absence of exogenous reducing agents, or whether chemical oxidation proceeds unimpeded through unstrained crystalline domains.

Crystallization control and buried substrate interfaces similarly divide current processing approaches [7,82,83]. Regulating the buried interface and crystal growth with SATS produces a 23.85% single-junction Sn-Pb PCE and an uncertified 28.74% 2T tandem PCE [82]. Substrate hydrophobicity imposes distinct constraints: on poly(triarylamine) (PTAA), non-wetting impedes uniform nucleation, and Lewis base additives such as 1-benzoyl-3-phenylurea (BPU) or its sulfonyl derivative (BPSU) assemble precursor clusters that accelerate nucleation, delay crystal growth, and passivate buried interface defects. These additives yield a single-junction Sn-Pb PCE of 23.87% on PTAA, a certified tandem PCE of 27.17%, and an uncertified champion of 23.87% [7]. Heteroepitaxial seeding presents an alternative growth route: in situ reaction of potassium stannate with lead iodide forms lead stannate (PbSnO₃), an ABX₃-structured oxide providing a 98% lattice match [83]. However, while PbSnO₃ templates achieve an uncertified champion 2T tandem PCE of 28.81%, the stabilized efficiency is only 23.12%. Epitaxial matching therefore does not overcome severe transient losses or interfacial charge accumulation [83].

Subcell current matching requires pairing the Sn-Pb absorber with a wide-bandgap (WBG) top cell absorbing between 1.75 and 1.80 eV [84,85]. A major debate centers on whether severe WBG VOC deficits arise from bulk halide phase segregation under illumination or from defect-mediated losses at contacts [2,14,86,87]. Supporting bulk segregation mitigation, the organic dye D358 reduces I₂ to I⁻ and oxidizes metallic Pb⁰ to Pb²⁺. Suppressing phase segregation raises WBG PCE from 18.75% to 19.94% and delivers an uncertified 28.83% tandem under International Summit on Organic Photovoltaic Stability light soaking (ISOS-L-1) [86]. Multifunctional cross-linkers also suppress phase segregation and nonradiative recombination, yielding a certified WBG VOC of 1.37 V and an uncertified 28.3% tandem PCE [87]. In MA-free compositions, 1,3,5-trifluoro-2,4,6-triiodobenzene (TFBZ) forms N-H...F hydrogen bonds with formamidinium iodide (FAI). Those bonds retard crystallization and passivate iodine vacancies and undercoordinated Pb²⁺ sites, achieving a certified tandem PCE of 28.52% (champion 29.01%) [2]. In parallel, an ether-ring supramolecule suppresses light-induced halide segregation, enabling 1.77 eV cells to reach 21.01% PCE, retain 95% of initial efficiency over 1000 h of MPPT, and reach a certified tandem PCE of 27.92% (champion 28.44%) [14].

In contrast to bulk segregation theories, substantial evidence attributes WBG open-circuit voltage deficits to unpassivated defects and dipole pinning at the buried hole extraction interface [88,89,90,91,92,93]. Sulfonic molecular bridges using sulfopropyl methacrylate (SPM) generate vertically aligned dipoles that accelerate charge extraction. The bridged device achieves a 1.77 eV WBG PCE of 19.48% with a VOC of 1.350 V, a VOC deficit of 0.420 V, and an uncertified 28.9% tandem PCE [88]. POEAX parallel dipoles similarly adjust energy alignment, yielding a 19.54% WBG PCE and an uncertified 28.92% tandem PCE [89]. Molecular bridging with 4F-HBM at the buried interface yields a 1.77 eV WBG device with 20.09% PCE and an 84.71% FF, exceeding control values of 18.47% and 82.53%. That device retains 85% of initial efficiency after 821 h of MPPT and supports a four-terminal (4T) tandem PCE of 28.71% [90]. Contact potential homogenization uses 3-aminopropanoic acid (3-APA) blended with [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (Me-4PACz) in a self-assembled monolayer (SAM). The blend delivers a VOC x FF product reaching 84.5% of the Shockley-Queisser (SQ) limit, achieving a champion p-i-n PCE of 22.4% [91]. In situ 5-ammonium valeric acid iodide (5-AVAI) molecular compensation repairs SAM-deficient

interfaces and grain-boundary defects [93]. Interfacial chemistry also dictates chemical stability. Strongly acidic phosphonic acid self-assembled monolayers (SAMs) corrode underlying nickel oxide (NiO_x), whereas less-acidic boric acid anchoring eliminates NiO_x degradation to form aggregation-free monolayers, enabling an uncertified 28.5% tandem PCE [92]. Transitioning to scalable deposition highlights phase management. Blade-coating 1.77 eV absorbers induces a parasitic 2H hexagonal phase, but adding 4-aminobenzylphosphonic acid suppresses the 2H phase to promote direct 3C cubic crystallization. That coating achieves a certified WBG PCE of 19.72% (champion 20.35%) and a 4T tandem PCE of 28.46%, though on a 0.07 cm^2 aperture [94].

Monolithic 2T integration depends on an interconnecting layer (ICL) that pairs current matching with negligible optical parasitism and low resistance [95,96,97,98,99]. One route replaces conventional Au/poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) stacks with a graphene oxide layer supporting [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz). The swap eliminates parasitic absorption and raises champion tandem PCE from 19.7% to 23.4% [97]. Omitting atomic layer deposition (ALD) and noble metals with a simplified recombination layer produces an uncertified 26.56% 2T tandem [98]. Controlling growth to expose binding-favourable facets suppresses interfacial recombination to enhance voltage [99]. Although monolithic all-perovskite tandems operate toward the two-junction detailed-balance limit of 47.6% [95], experimental certified efficiencies remain between 27.17% and 29.2% on laboratory-scale cells [2,6,7,8,9,10,14]. Crucially, scaling to practical dimensions incurs severe penalties. Surface treatment with propane-1,3-diammonium iodide yields a 1.77 eV subcell VOC of 1.35 V and a PCE of 19.9%, but scaling the tandem to 1 cm^2 reduces the champion PCE to 26.0% (at a 17.8% solar-to-hydrogen efficiency) [100]. Theoretical projections reach 27.3% for dual-junction $\text{KSnI}_3/\text{FASnI}_3$, and 30.69% PCE with a VOC of 2.766 V and an FF of 84.18% in triple-junction $\text{KSnI}_3/\text{FASnI}_3/\text{ACZTSe}$ tandems [101], but these projections rely on idealized transport models that neglect the chemical oxidation, phase segregation, and sheet resistance observed in practical devices.

5. Hybrid Tandems: Coupling Perovskite to Silicon, CIGS and Organic Subcells

Wide-bandgap (WBG) perovskite top cells coupled to narrow-bandgap bottom absorbers suffer from photoinduced phase segregation. Studies across the year reach incompatible conclusions regarding the governing physical mechanism. One camp contends that crystallographic orientation dictates halide demixing by modifying ion-migration energy barriers. Trioctylphosphine oxide additives induce preferential (111) texturing, reported to elevate ion-migration activation energy and suppress phase separation relative to the (100) facet, yielding a certified two-terminal (2T) power conversion efficiency (PCE) of 30.26% (champion 30.78%) on textured silicon, with stability exceeding 1,000 hours [13]. In contrast, bidentate anchoring with methylammonium chloride derivatives (MAMCl) directs preferential (100) nucleation, asserting instead that (100) orientation relieves lattice strain to suppress non-radiative defects in 1.77 eV absorbers. Its tandem achieved a certified 2T tandem PCE of 28.06% (champion 29.0%) with an open-circuit voltage (Voc) of 1.369 V and 20.4% single-junction efficiency [4]. Rejecting crystallographic texture as secondary, competing reports argue that phase stability is governed by chemical alloying or ionic site competition. Melamine additives in rubidium-alloyed lattices suppress phase separation and enhance carrier mobility, yielding a 1.68 eV cell with 25% PCE, 1.31 V Voc, and 86.4% fill factor (FF). That cell enabled an uncertified champion 2T perovskite/silicon tandem PCE of 33.5% (stabilised at 33.5%) with an International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-2 T80 lifetime of 3,200 hours [102]. Concurrently, ionic site competition directs Cl^- anions into lattice rather than interstitial sites, raising single-junction

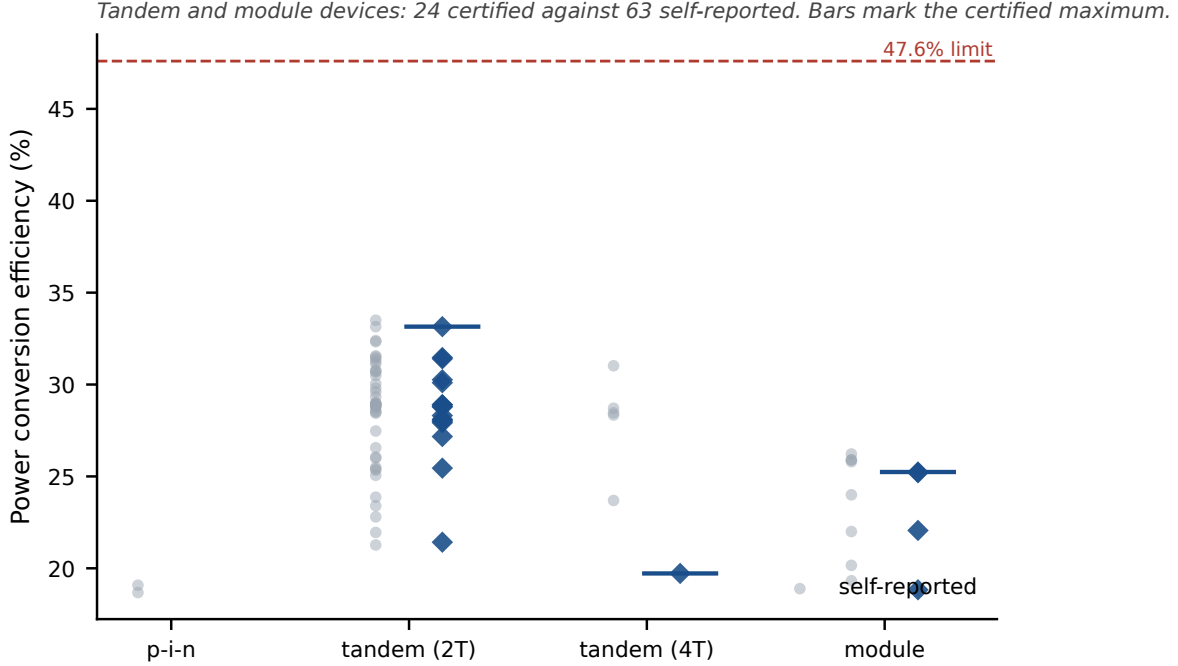


Figure 4. Tandem and module devices: independently certified against self-reported champion efficiencies. The dashed line marks the 47.6 percent two-junction detailed-balance limit. Simulation and review studies are excluded.

WBG efficiency to 23.88% and 2T tandem efficiency to 32.39% under continuous testing [103]. Settling whether facet orientation, cation alloying, or interstitial occupancy governs phase segregation requires operando, depth-resolved grazing-incidence wide-angle X-ray scattering. That measurement must be paired with nanoscale secondary-ion mass spectrometry under 1-sun bias to decouple surface recrystallization from bulk halide redistribution.

Contradictory evidence also surrounds substrate morphology and residual mechanical stress in perovskite/copper indium gallium selenide (CIGS) tandems. Conventional wisdom assumed planarizing rough chalcopyrite was required to avoid shunts. However, polished CIGS substrates induce harmful tensile stress in perovskite films that weakens bonding and accelerates degradation [15]. Retaining corrugated, rough CIGS topographies mechanically relaxes tensile stress, securing a certified 28% efficiency (stabilised at 28%) and an operational T80 of 1,123 hours [15]. Conversely, flexible monolithic perovskite/CIGS tandems on polymer substrates achieve 22.8% PCE with Voc exceeding 1.8 V on smooth substrates. That result follows from modifying radio-frequency sputtered nickel oxide (NiO_x) with [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid derivatives (4PADCB) to form $\text{NiO}_x\text{-P}$, which suppresses interface recombination without substrate corrugation [104]. In perovskite/silicon tandems, localized silicon oxide (SiO_x) nanosphere contacts sit on industrial silicon with iceberg-like pyramids exceeding $2\text{ }\mu\text{m}$, and those contacts permit complete solution coverage and mitigate interfacial recombination, delivering a certified 33.15% tandem PCE [11]. Vapor-solution hybrid two-step crystallization modulation similarly enables conformal WBG growth on fully textured silicon for a certified 31.4% PCE [12]. Meanwhile, 5C7A additives relieve internal stress and elevate the ion migration barrier, raising inverted single-junction PCE to 23.25% and textured silicon tandem PCE to 30.59% [105].

Mechanistic divergence extends to hole-selective contacts based on self-assembled monolayers (SAMs), where one model asserts that maximizing permanent molecular dipoles drives rapid charge extraction through enhanced band bending. The Knoevenagel-condensed molecule BCPA possesses an electrostatic dipole moment of 18.16 Debye, far exceeding the 1.60 Debye of [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), and that molecule raised single-junction efficiency to 19.08% and perovskite/organic tandem efficiency to 26.10% [106]. Asymmetric fluorinated SAMs with enlarged dipoles similarly boost perovskite/passivated emitter and rear cell (PERC) tandems beyond 30% efficiency with a 30.05% champion PCE [107]. Conversely, competing evidence indicates that anchoring uniformity outweighs dipole magnitude. Blending 4-aminobenzylphosphonic acid with Me-4PACz constructs a synergistic self-assembled monolayer (SAM) that reinforces buried anchoring, yielding an inverted single-junction certified PCE of 25.45% (champion 25.75%) and a 2T tandem PCE of 31.56% on 1.07 cm^2 [17]. Bifacial reinforcement via 4-PhCz improves SAM coverage on indium tin oxide (ITO), and the 1.67 eV single junction reached a Voc of 1.273 V with a voltage loss of 0.397 V (22.53% PCE), alongside a 31.26% 2T tandem [108]. Super-hydrophilic chlorine-substituted molecules deliver 23.97% single-junction and 31.12% tandem efficiencies via enhanced wetting [109]. Micelle-assisted SAM deposition yields 20.95% single-junction and 29.8% tandem PCE on 1 cm^2 , retaining 92% performance after 300 hours of maximum power point (MPP) tracking under unfiltered Xenon illumination [110]. Inorganic interlayers further modulate SAM interfaces: Ti₃C₂T_x MXene nanosheets form Ti-O-P bonds that raise the MeO-2PACz work function [111], and Nb-MXene paired with S-benzyl-L-cysteine reduces valence band offsets [112]. TbCl₃ doping at the Me-4PACz/CsPbI₃ junction boosts all-inorganic single junctions to 18.68% and silicon-based tandems to 29.40% in four-terminal (4T) and 25.44% in 2T configurations [113]. Multi-site agents passivating NiO_x, the buried interface, and silicon edges simultaneously achieve a 31.02% 4T tandem [114].

At the interconnecting layer (ICL) coupling subcells, optical parasitism conflicts with ohmic carrier recombination across every tandem chemistry surveyed. In perovskite/CIGS tandems, an Al:ZnO/Au/NiO_x/4PADCB composite ICL uses an ultrathin Au layer to accelerate recombination and a hybrid NiO_x/4PADCB hole transport layer (hole-transport layer (HTL)) to align energy levels, attaining a certified 30.1% PCE (champion 30.71% on 0.15 cm^2) [16]. In contrast, BMIMTFSI ionic liquid integration into SAMs achieves 28.34% in 4T perovskite/CIGS tandems (22.53% WBG and 25.68% normal-bandgap cells) [115]. In lead-free tandems combining tin perovskites with CIGS, Pd(dppf)Cl₂ functionalization delivers an 11.52% single-junction and 14.51% tandem PCE [116]. Dual-interface passivation with succinic acid at SnO₂ and monoethyl succinate at copper indium disulfide (CIS) subcells instead reduces diode ideality factors from 2.25 to 1.26 to yield 18.76% PCE [117]. In perovskite/organic tandems, tailoring Au nanoparticle dimensions in n-i-p ICLs suppresses plasmonic parasitic absorption, providing a current gain exceeding 1.5 mA cm^{-2} in the organic rear cell and a 25.34% tandem PCE [118]. Alternatively, metal-free SAM/MoO₃/SAM junctions convert the molybdenum oxide surface from n-type to p-type to yield 26.05% PCE [119]. Similarly, 1,2,4-trichlorobenzene additives enforce face-on SAM packing for 26% tandems [120], while sputtered zinc tin oxide (ZTO) provides an indium-free recombination layer [121]. Operational degradation in perovskite/organic devices arises from C₆₀ electron transport layer (electron-transport layer (ETL)) recrystallization, which opens grain boundaries to halide and silver interdiffusion. Organic barrier layers suppress this corrosion to sustain 25.5% efficiency [122]. Polymerizable additives in 1.86 eV WBG absorbers retain 84% of initial efficiency after 1,000 hours of maximum power point tracking (MPPT) in 25.06% tandems [123]. Dual passivation with MESK and octylammonium iodide yields 23.69% in 4T tandems [124]. Interface modeling indicates that PbI-terminated

MAPbI₃/poly(3-hexylthiophene) (P3HT) heterojunctions provide stronger orbital hybridization and valence band alignment than MAI terminations [125].

Top-interface passivation and module scaling introduce further mechanistic trade-offs, beginning with cation competition at the perovskite surface. Pairing 1,3-propanediamine with trifluoroacetate counterions in PDA(TFA)₂ suppresses that competition, delivering 23.23% in 1.68 eV single-junction cells with 1.27 V Voc, and the same route gave a certified 31.47% tandem PCE (champion 32.33% on 0.945 cm²) [5]. Bidirectional ion transport from piperazinium chloride passivates top and buried interfaces simultaneously, reaching 22.3% in single junctions with a Voc × FF product at 84.4% of the Shockley-Queisser limit. The same passivator gave 31.5% in 2T tandems on 1.04 cm² with a T80 of 755 hours [126]. Cooperative dipole reconstruction with acetylcholine chloride and n-octylammonium iodide raises Voc to 1.29 V, capturing 92.8% of the Shockley-Queisser voltage limit in a 21.27% tandem [127]. In 3D/quasi-2D Dion-Jacobson heterostructures, combined evaporation and solution deposition yields a certified 20.53% single-junction PCE (champion 20.71%) and a certified 28.81% tandem PCE (champion 28.99%) with 501 hours T80 [1]. Across large areas, combining phosphonic acids with piperazinium chloride increases quasi-Fermi level splitting by approximately 100 mV, enabling certified 28.9% efficiency on 60 cm² [18]. VBETS passivators instead deliver 30.98% tandems and 21.00% across 32.144 cm² modules [128]. At silicon contacts, oblique-angle evaporation deposits conformal SiO_x electron contacts without damaging underlying perovskite in tunnel oxide passivated contact (TOPCon) tandems [129]. Ultrathin poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) sacrificial layers instead prevent resistive SiO_x growth during annealing, maintaining a 0.813 FF in 21.95% tandems [130]. Sulfonium DMPESI molecules form internal hydrophobic encapsulation layers that suppress degradation during air-annealing, and those devices retained 84% efficiency after nearly 1,800 hours of MPP tracking under ISOS-L-1 in 30.49% tandems on 1.21 cm² (T80 of 723 hours) [131]. MXene grain boundary additives dissipate heat to yield 25.13% PCE with 500 hours T80 [132]. Certified single-junction perovskite efficiency reached 27.3%, and hybrid perovskite/silicon tandems approach 35% in small-area cells [133]. These values remain substantially below the theoretical detailed-balance limit of 47.6% for two-junction tandems. Simulated annual energy yields scale identically with top-cell bandgap across geographies, indicating that a twofold relaxation in perovskite degradation rates is sufficient to match silicon leveled cost of electricity [134].

6. Modules and Large-Area Devices: Scaling the Deposited Film

A sharp mechanistic division separates scalable perovskite deposition reports: at issue is whether film heterogeneity across large apertures is governed by fluid-phase drying and solvate-evaporation kinetics, or pre-emptively dictated by nanoscale molecular aggregation at the buried substrate. One body of work attributes macroscopic defect striations during blade or slot-die translation to convective transport and slow desolvation of high-boiling solvents [135,136,137]. The opposing body asserts that liquid drying reflects underlying contact topography. On that view, self-assembled monolayer (SAM) dewetting, micelle aggregation, and weak interfacial anchoring create lateral nucleation gradients that solvent tuning cannot resolve [138,139,140]. This dispute persists because fluid-drying studies evaluate uncertified small-area cells (<0.1 cm²) without tracking lateral crystallization kinetics, whereas buried-contact studies rarely quantify ink shear forces during dynamic translation. Whether bulk evaporation or interfacial adhesion governs large-area film quality is therefore unresolved, and settling it requires synchronous operando grazing-incidence wide-angle X-ray scattering (GIWAXS) and hyperspectral photoluminescence (PL) mapping across translating ink menisci under

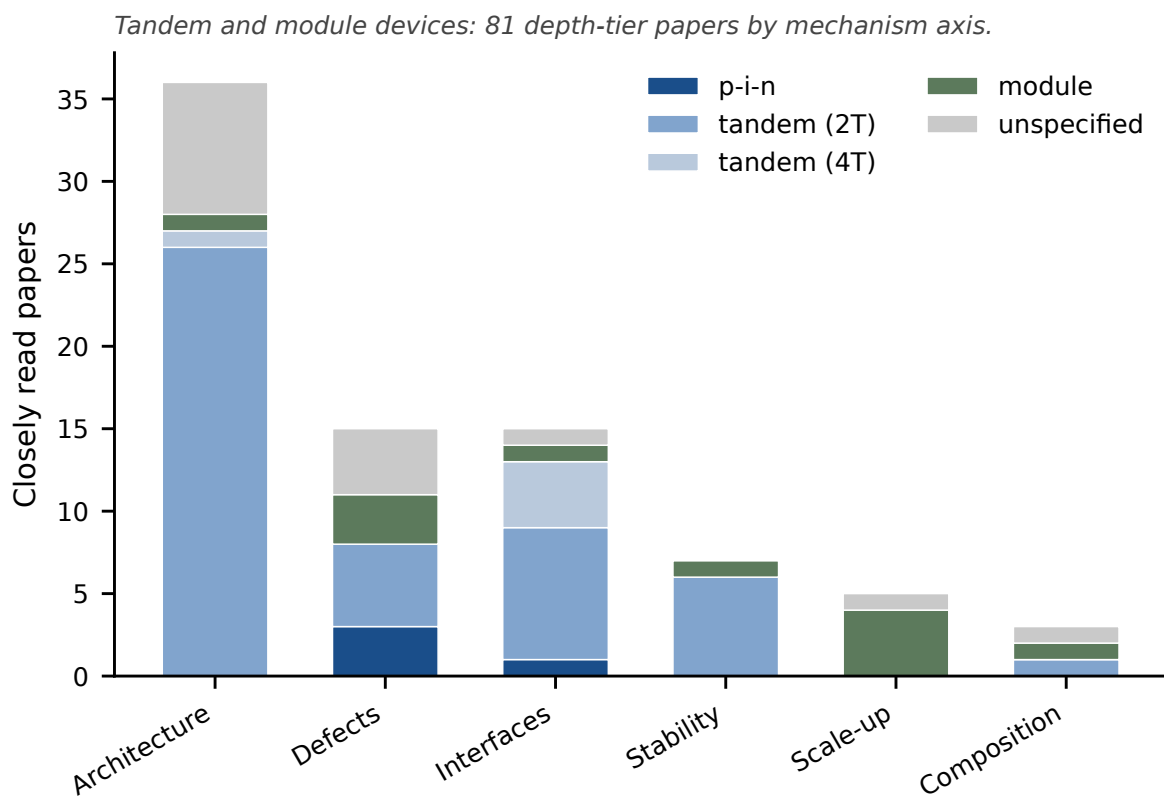


Figure 5. Tandem and module devices: where the year’s close reading was directed, by mechanism axis and device architecture.

controlled shear rates, coupled to local adhesion profiling across decimetre substrates.

Within fluid-phase processing, conflicting strategies govern intermediate coordination complexes during the liquid-to-solid transition. One approach insists high-boiling polar solvents such as N-methyl-2-pyrrolidone (NMP) must be expelled rapidly to avert defective solvates. Propionic acid weakens lead iodide (PbI_2)-NMP coordination to accelerate evaporation in antisolvent-free coating, yielding an uncertified champion power conversion efficiency (PCE) of 25.86% on small cells [135]. Volatile pyridine coordination purges solvent residues in narrow-bandgap tin-lead mini-modules to reach $22.0 \pm 0.4\%$ PCE across 10.4 cm^2 [141]. Conversely, delaying desolvation is argued to ensure large-area uniformity. A ternary 2-methoxyethanol (2ME), dimethylformamide (DMF), and NMP system preserves intermediate coordination to retard premature crystallization, achieving a certified 18.73% PCE across 100.15 cm^2 with 95.8% retention after 1,0 h at maximum power point (MPP) [136]. Thermal input presents an identical dilemma: traditional furnace annealing generates convective thermal gradients that precipitate secondary phases, whereas rapid laser annealing at 20 W cm^{-2} establishes a $123 \pm 18 \text{ s}$ degradation-free ambient processing window that suppresses 6H hexagonal phase accumulation, reaching 24.0% PCE over 100 cm^2 [137].

Additive sequencing and phase restoration present parallel conflicts between pre-deposition ink equilibria and post-deposition crystallization. Incorporating hexamethylphosphoramide (HMP) inverts the reaction sequence between formamidinium (FA^+) and rubidium (Rb^+) with the Pb-I framework during ambient blade coating (relative humidity (RH) $\leq 55\%$), producing 25.8% PCE on an uncertified 0.09 cm^2 cell [142]. L-ascorbic acid sodium salt (L-AASS) complexation with PbI_2 suppresses residual halide phases to deliver 25.23% PCE on a 65 cm^2 module [143]. However, relying on precursor additives risks phase segregation in multi-cation inks. In situ vapor post-treatment using formamidinium iodide (FAI) and methylammonium thiocyanate (MASCN) bypasses solution equilibria by converting residual PbI_2 and delta-phase FAPbI_3 into photoactive $\alpha\text{-FAPbI}_3$. That treatment reaches a certified 25.67% PCE (champion 26.05%, stabilized 25.41%) on small areas, 24.52% on 1 cm^2 , and 22.35% on a 17.1 cm^2 mini-module [144]. Compositional design during scalable coating shows that eliminating methylammonium (MA^+) via formamidinium-cesium alloys (FACsPbI_3) prevents volatile organic degradation [145]. Incorporating chlorine through PbCl_2 or perovskite chloride additives into slot-die coated triple-halide wide-bandgap formulations widens the optical bandgap by tens of millielectronvolts, yet yields only 17.2% PCE on an uncertified 0.062 cm^2 aperture [146].

Interfacial defect arguments attribute scaling losses to morphological instabilities of hole-selective self-assembled monolayers (SAMs) on metal oxides. As-prepared SAM solutions assemble into $\sim 460 \text{ nm}$ fibrous micellar aggregates that seed pinholes and potential fluctuations. Dispersing them with dipolar additives yields monodisperse layers, achieving an uncertified champion PCE of 26.58% (certified 25.81%) on 0.09 cm^2 with T90 extrapolated to 11,259 h under International Summit on Organic Photovoltaics Stability (International Summit on Organic Photovoltaic Stability (ISOS)) ISOS-D-1 [138]. In continuous blade coating over $10 \text{ cm} \times 10 \text{ cm}$ substrates, 3-nitrobenzenesulfonic acid (3NS) co-adsorption enforces uniform SAM anchoring, yielding 20.16% PCE across a 63.09 cm^2 module [139]. Co-assembly strategies similarly suppress aggregation: co-adsorbing salicylic acid (SA) with [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) fills pinholes to yield a certified 28.78% PCE (champion 28.94%) with 1.332 V open-circuit voltage (VOC) in a 0.087 cm^2 wide-bandgap tandem cell [3]. Co-assembling the co-adsorbent PDADI with [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) avoids segregation to deliver a certified 22.09% PCE (champion 25.49%) on 8.5 cm^2 [140]. Reinforced undecylphosphonic acid (UPA) anchors opti-

mize energy alignment to attain 25.9% PCE [147]. Solid-vapor impregnation of the phosphonic acid derivative 4PADCB with glycine hydrochloride curbs SAM sublimation, yielding 19.18% PCE on a 5 cm x 5 cm module and 22.15% on 0.16 cm² [148].

Interfacial mechanics further divides the field: chemical defect passivation fails if mechanical delamination occurs under thermal and scaling stress. In situ crosslinking of hexamethylene diisocyanate (HDI) from the perovskite ink with ethylene glycol (EG) on the electron transport layer (electron-transport layer (ETL)) fastens the buried contact and relaxes tensile strain. The crosslinked stack yields 20.40% PCE on a 25 cm² module alongside 26.04% on rigid and 24.16% on flexible small devices [149]. Potassium benzyl(trifluoro)borate (BnBF₃K) linkers similarly enhance adhesive toughness at the tin oxide (SnO₂)/perovskite interface, producing a certified 21.39% PCE (champion 24.15%) [150]. Bilateral bond strength equilibrium through 2-(benzo[d]thiazol-2-ylthio)acetic acid (BTSA) on nickel oxide (NiO_x) anchors sulfur, nitrogen, and aromatic rings, enabling a certified 26.65% PCE (champion 26.98%) across a 764 cm² module [25]. In n-i-p modules, anchoring SnO₂ with methylglycinediacetic acid trisodium salt (MGDA · 3Na) yields 23.27% PCE across 5 cm x 5 cm (26.43% on small cells) [151]. Formamidinium salts at the buried interface instead assist charge extraction, reaching 21.72% PCE on an 18.55 cm² mini-module (25.61% on single devices) [152]. Dual interface modification with Nd₂O₃ on SnO₂ curbs lattice mismatch and ion migration to maintain 89% retention after 2,0 h continuous illumination, though restricted to 0.045 cm² cells at 26.22% PCE [153].

Module-level interconnects and transport layers introduce distinct failure modes. Slot-die coating of coordinated hole transport layers (HTLs) preserves conductivity across expanded areas, delivering 22.80% PCE at 23.25 cm², 21.45% at 87.45 cm², and 20.21% at 749.276 cm² (certified 19.5% against a 26.02% cell at 0.0655 cm²). Those modules also pass module quality test (MQT) standards MQT 08, MQT 10, and MQT 13 under International Electrotechnical Commission (IEC) standard IEC 61215-2-2021 [154]. Replacing hygroscopic lithium salts in spiro matrices with spiro-phenothiazine derivatives (PTZ-FI) immobilizes mobile ions. That replacement delivers 22.1% PCE on a 5 cm x 5 cm mini-module (certified 25.2%, champion 25.8% on small cells) with >85% efficiency retained after 1,100 h under ISOS-D-1 [19]. Fluorene-terminated spiro p-BM achieves 22.4% PCE on a 16 cm² mini-module (25.5% on 0.0625 cm²) [155]. Light-accelerated oxidation of 2,2',7,7'-tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (spiro-OMeTAD) via ammonium bis(trifluoromethanesulfonyl)imide (TFSI) salts eliminates mobile metal cations, yielding a certified 20.95% PCE on a 12.83 cm² module with 7% degradation after 700 h unencapsulated operation [156].

Top interfaces and grain boundaries govern remaining module voltage shortfalls. In situ grain boundary polymerization with dimethyl itaconate eliminates inter-grain gaps, enabling 30 cm x 30 cm modules to achieve 19.2% PCE (from an uncertified 25.9% small cell, certified at 25.24% on 0.09 cm²) [157]. Conjugated ionic additives in poly(triarylamine) (PTAA) passivate boundary traps to yield 22.96% PCE at 15.17 cm² (champion cell 25.73%) [158]. Entropy-driven stabilization using 2-naphthalenethiol derivatives (2NTD) inhibits polyiodide defect formation to yield a certified 26.4% PCE (champion 26.63%) on 0.09 cm² [159]. In situ grown two-dimensional (2D) CHEA2PbI₄ capping layers on cesium lead triiodide (CsPbI₃) absorbers align interfacial energy levels, delivering a certified 18.83% PCE (champion 19.32%) on a 12.44 cm² mini-module [160]. Wafer-scale monolayer molybdenum disulfide (MoS₂) buffers physically impede halide migration into transport contacts, retaining <5% PCE loss after 1,200 h damp heat (85 deg C and 85% RH) on 0.074 cm² cells (certified

25.9%, champion 26.2%) [161]. Radical molecular network-buffers limit VOC deficits to an energy loss of 0.297 eV, attaining 21.39% PCE on a 12.5 cm² mini-module (25.28% on small cells) [162]. Expanding chlorine-doped FAPbI₃ absorbers on ultraviolet-insensitive SnO₂ to a 16 cm² photoelectrode platform demonstrates 140 h operation. That result confirms that large-area stability depends upon eliminating unreacted boundary defects rather than merely enlarging film dimensions [163].

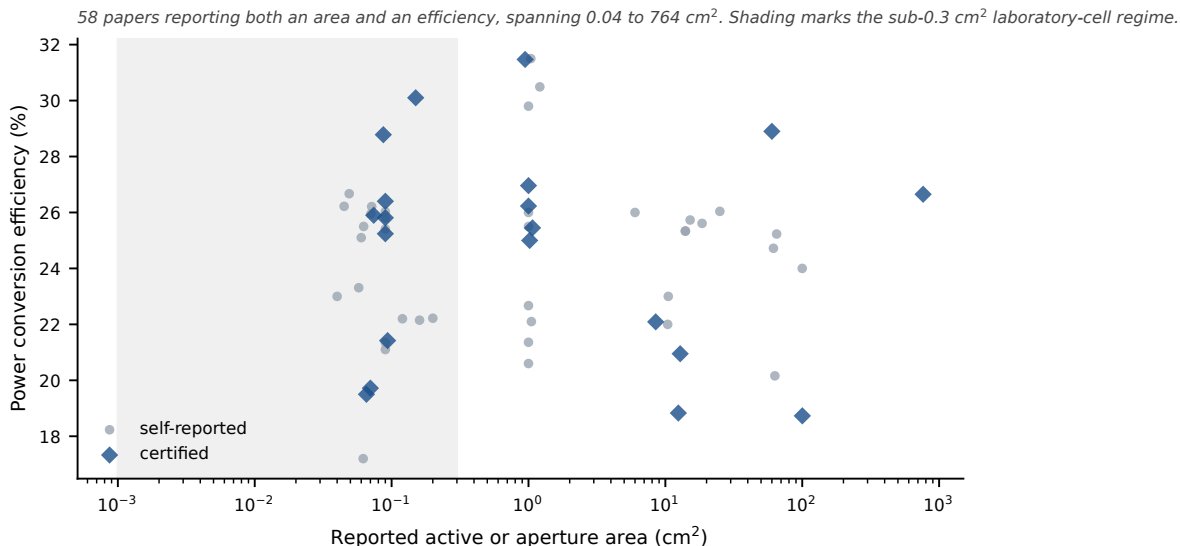


Figure 6. Efficiency against reported active or aperture area. Shading marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

7. Reading the Device Classes Together

Reading the device classes side by side reveals that performance gains claimed in one architecture are frequently paid for through degradation pathways detailed in another. In single-junction cells and wide-bandgap (WBG) subcells, carbazole-based self-assembled monolayer (SAM) hole transport layers (HTLs) optimize interfacial dipoles and passivate buried substrate traps. Those contacts yielded a certified two-terminal (2T) tandem power conversion efficiency (PCE) of 28.78% [3] and 31.47% on 0.945 cm² [5]. However, reading these contact modifications alongside large-area module manufacturing demonstrates a conflict: the molecular characteristics responsible for low interfacial recombination velocity directly impede scalable processing. The hydrophobic end groups and dipole moments of phosphonic acid self-assembled monolayers (SAMs) form micellar agglomerates of roughly 460 nm that cause severe dewetting during solution casting [138]. When translated from spin-coating to continuous blade coating across large substrates, these aggregates produce pinholes and non-uniform nucleation. Co-assembly strategies are then required, and they yield 20.16% on a 63.09 cm² module [139] or 22.09% certified on 8.5 cm² [140]. In addition, the redox susceptibility of SAM anchors on nickel oxide under operational bias creates an interfacial failure pathway. The high open-circuit voltage (Voc) demonstrated on small-area devices is purchased through mechanical and chemical fragility that drives rapid delamination under lateral hydrodynamic shear during scalable coating.

A parallel conflict appears at top interfaces, where surface passivators that optimize single-junction cells generate transport bottlenecks in monolithic tandems. Single-junction devices employ bulky organic cations to form two-dimensional (2D) or quasi-2D capping heterojunctions that passivate halide vacancies and suppress non-radiative decay. Those caps delivered a certified single-junction

PCE of 26.96% with a champion of 27.02% and a stabilized output of 26.61% on 1 cm^2 under continuous 1-sun maximum power point (MPP) tracking [22], alongside 28.81% certified in 2T tandems [1]. Yet when integrated into monolithic tandem stacks, the low out-of-plane conductivity and wide bandgap of these low-dimensional capping layers establish electrostatic potential steps and transport barriers. Under operational bias, these barriers trap photogenerated carriers and create electrostatic crosstalk across the interconnecting layer (ICL). In addition, transparent conducting oxides sputtered directly onto low-dimensional capping layers induce chemical and mechanical damage. Unshielded operational electric fields separately lower the activation energy for halide migration beneath the passivating cap. What single-junction studies classify as defect deactivation functions in tandem stacks as an extraction barrier that accelerates phase demixing under continuous operational load.

Comparing narrow-bandgap (NBG) mixed tin-lead (Sn-Pb) subcells with WBG mixed-halide top cells reveals that two separate literatures describe identical degradation physics under distinct vocabularies. The Sn-Pb literature attributes operational decay to chemical Sn(II) oxidation. Reducing agents such as tris(2-carboxyethyl)phosphine (TCEP) achieve 23.86% single-junction efficiency, 90% retention after 563 h of maximum power point tracking (MPPT), and an uncertified 28.86% 2T tandem champion [81]. Hydrazine monohydrochloride instead achieves 23.21% low-bandgap single-junction PCE and 28.31% certified 2T tandem PCE [6]. Concurrently, WBG top-cell studies treat halide demixing as a light-induced phase separation problem governed by crystallographic orientation. That literature utilizes (111) texturing to achieve a certified 2T tandem PCE of 30.26% with over 1,0 h stability [13], or bidentate anchoring via methylammonium chloride derivatives (MAMCl) for (100) orientation to yield 28.06% certified 2T tandem PCE [4]. Reading these literatures together reveals that local lattice microstrain is the primary driver in both subcells. Incorporating sodium 1,3,6-naphthalenetrisulfonate (NTS) or lattice-stabilizing networks into Sn-Pb lattices homogenizes microstrain. Those routes achieve certified 2T tandem efficiencies of 28.9% [8] and 29.2% [10], demonstrating that Sn(II) oxidation is secondary to vacancy formation caused by lattice distortion. In both architectures, unrelaxed microstrain lowers the kinetic activation barrier for ion migration: the same strain that permits phase segregation in mixed-halide top cells accelerates vacancy-assisted oxidation in mixed-metal bottom cells. However, monolithic 2T integration prevents applying both chemical remedies concurrently, as volatile reducing additives in the Sn-Pb subcell degrade during high-temperature deposition of subsequent layers.

Synthesizing findings across all device classes demonstrates that the year’s evidence does not permit simultaneous claims of high certified efficiency, scalable area, and verified operational durability. The highest efficiency benchmarks are confined to small, uncertified apertures. One 2T perovskite/silicon tandem reached an uncertified champion PCE of 33.5% with an International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-2 time to 80% of initial performance (T80) lifetime of 3,200 hours [102], and an uncertified 25.86% cell was prepared via solvent coordination [135]. Scaling to large areas causes efficiencies to drop to 18.73% certified on 100.15 cm^2 [136] or 24.0% uncertified on 100 cm^2 [137], without accredited operational lifespans. Meanwhile, durable devices such as perovskite/copper indium gallium selenide (CIGS) tandems achieving 1,123 h T80 at 28% certified PCE [15] require compliant interlayers that compromise absolute efficiency. Molecular passivators that suppress defects on static small cells cause dewetting during blade coating and electronic crosstalk across tandem interconnections. Gains reported within isolated device classes therefore cannot be combined additively.

Table 2. The evidence base behind each section. Extracted values are individual numbers bound

to a source quotation; verified quotations are anchors confirmed word for word against the source abstract by a script sharing no code with the extraction stage.

Device class	Papers		Simulation or review	Extracted values	Verified quotations
	read	Measured			
Single-junction cells	332	278	54	308	1175
All-perovskite tandems	28	24	4	33	105
Perovskite/silicon and other hybrid tandems	47	44	3	62	185
Modules and large-area devices	43	41	2	76	190

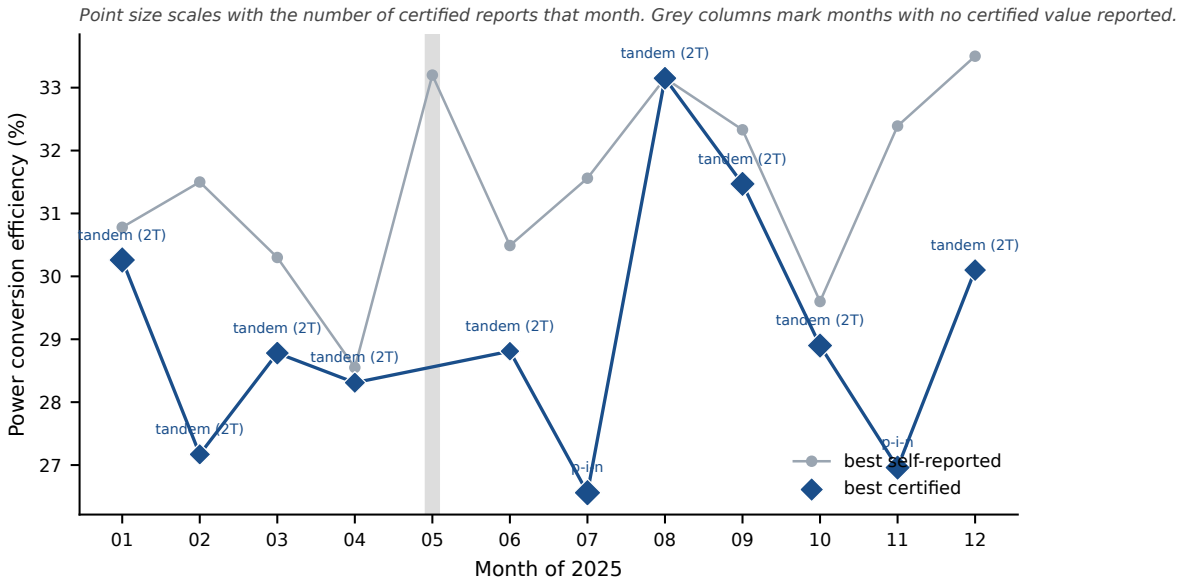


Figure 7. Best independently certified efficiency in each month of the year, with the device class of the leading value at each point. Point size scales with the number of certified reports.

8. Reporting Practice Across 2025

Automated regular-expression detection over abstracts identifies what authors state in title-level summaries rather than verifying experimental execution within full texts. Across 4730 abstracts, the detection rate for power conversion efficiency (PCE) was 51.6% pooled over summed denominators, remaining flat across the year with a narrow monthly range of 47.8% to 54.0%. This confirms that steady-state photovoltaic conversion serves as the standard self-reported metric. In contrast, the detection rate for independently certified efficiency was only 5.8% of 4730 abstracts, fluctuating between 3.4% and 7.6% across monthly cohorts. Such a restricted detection rate leaves champion claims unverified against potential spectral mismatch, stray-light artifacts, or calibration drift. Rigorous independent certification benchmarks high-performance devices, as demonstrated by certified values of 26.03% in single-junction cells [65], 25.18% on inverted contacts [164], and 28.31% in monolithic tandems [6], yet these verified results remain rare exceptions within the broader literature.

The gap between operational steady-state performance and dynamic current-voltage (J-V) sweeps is highlighted by the low detection rate for stabilised efficiency, which pooled at 8.8% of 4730 abstracts. Unlike the flat rate observed for raw efficiency, stabilised efficiency detection swung across the year from a minimum of 6.2% to a maximum of 12.0%, reflecting an accelerating but incomplete shift toward continuous maximum power point tracking (MPPT). Without stabilised power outputs, rapid voltage sweeps routinely record transient capacitive charge collection and mobile ion redistribution rather than electronic equilibrium. This diagnostic ambiguity is exacerbated by the near-total absence of hysteresis disclosures, detected in only 3.1% of 4730 abstracts with a flat monthly range of 1.6% to 4.3%. Because mobile halide vacancies accumulate at contact interfaces under external bias to create internal screening fields, unstated scan-rate-dependent hysteresis obscures carrier extraction kinetics. Where hysteresis is explicitly examined and resolved, it correlates directly with targeted defect elimination, such as dual-anchor interfacial passivation [165], organic potassium salt chelation [67], or B-site lattice doping that elevates migration barriers against mobile iodide drift [6,166].

Device active area was stated in only 7.3% of 4730 abstracts, remaining tightly bounded between 5.2% and 9.3% throughout the year. Omitting geometric dimensions across the overwhelming majority of abstracts conceals whether efficiency breakthroughs reflect scalable manufacturing or edge-enhanced collection on sub-0.1 cm² apertures. High certified efficiencies frequently concentrate on small active areas of 0.09 cm² [159,164]. When active area expands, non-uniform drying during meniscus coating, local shunt pinholes, and transparent conducting oxide sheet resistance induce severe geometric efficiency drops, decreasing from 26.40% on 0.09 cm² to 23.08% on a 12.96 cm² mini-module [159]. Although bilateral interfacial bonding and rapid laser annealing achieve certified efficiencies of 26.65% across 764 cm² [25] and 24.0% across 100 cm² [137], such large-area verifications remain exceptional amid widespread silence regarding cell geometry.

The thinnest reporting metric across the annual corpus was the International Summit on Organic Photovoltaic Stability (ISOS) protocol label, detected in merely 1.1% of 4730 abstracts. Although this detection rate ranged monthly from 0.2% to 3.2%, the pooled rate of 1.1% demonstrates that standardized aging methodologies remain virtually absent from high-level dissemination. Unstandardized degradation tests fail to distinguish extrinsic encapsulation failure from intrinsic lattice decomposition. In contrast, groups adhering to standardized protocols systematically isolate degradation pathways, deploying ISOS-L-1 continuous illumination tracking [57,68] or ISOS-L-2 thermal-light stress [102,164] to evaluate self-assembled monolayer (SAM) contacts [24,140]. Adopting standardized ISOS protocols alongside mandatory disclosure of aperture areas, stabilised tracking, and certified calibration is necessary to convert self-reported claims into reproducible scientific evidence.

Table 3. Reporting completeness by device class. Each cell gives the percentage of papers whose abstract states the quantity, with the paper count in brackets. Flags are recomputed by script from abstract text, so every figure is a lower bound: a paper reporting a quantity only in its experimental section is counted as not reporting it. No validated precision figure is available.

Device class	Certification stated	Area stated	Stabilised output	ISOS	
				protocol named	Papers read
Single-junction cells	9.0% (30)	7.8% (26)	1.2% (4)	5.1% (17)	332
All-perovskite tandems	28.6% (8)	7.1% (2)	3.6% (1)	3.6% (1)	28

Device class	Certification stated	Area stated	Stabilised output	ISOS protocol named	Papers read
Perovskite/silicon and other hybrid tandems	25.5% (12)	17.0% (8)	4.3% (2)	6.4% (3)	47
Modules and large-area devices	37.2% (16)	55.8% (24)	2.3% (1)	7.0% (3)	43

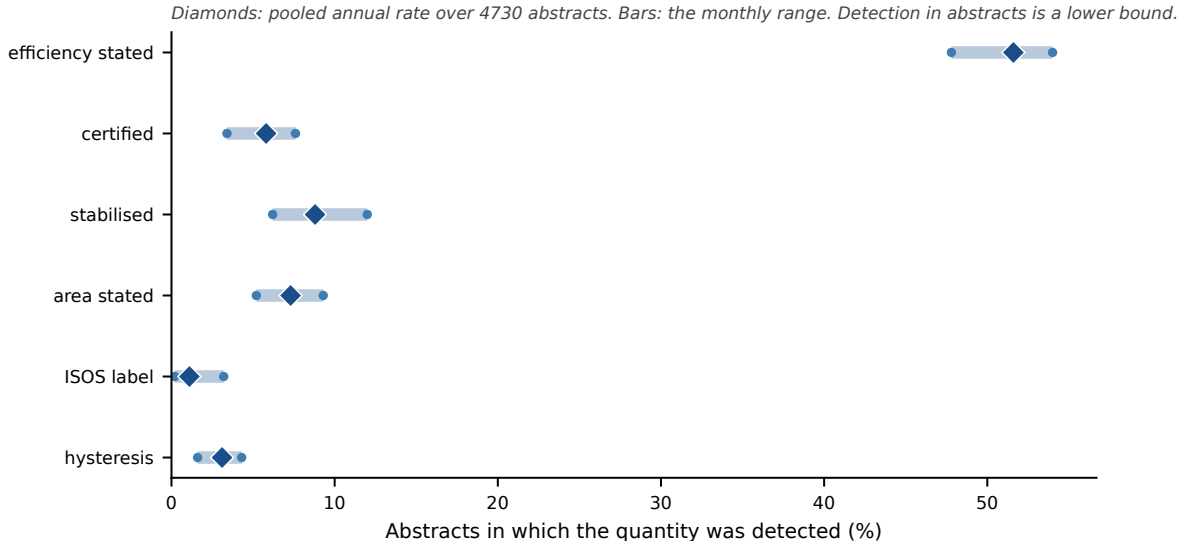


Figure 8. Reporting-practice detection rates across the year. Diamonds mark the pooled annual rate and bars the monthly range. Rates are computed over abstracts and measure what abstracts state, not what the underlying papers did.

9. Research Gaps and Outlook

Reconciling the 2025 literature exposes an empirical disconnect: champion efficiency records and verified operational durability remain mutually exclusive. Across 7931 works meeting the scope gate and 450 read closely, only 66 reported an independently certified power conversion efficiency (PCE), only 15 reported a time to 80% of initial efficiency (T80) lifetime, and only 24 named an International Summit on Organic Photovoltaic Stability (ISOS) protocol. The twelve-month certified frontier demonstrates punctuated dispersion rather than continuous progress, between 2025-01 (best certified 30.26% in two-terminal (2T) tandems [13]) and 2025-12 (best certified 30.1% [16]), the net change in best certified efficiency was -0.16 percentage points. Uncertified self-reported claims outstripped certified benchmarks in eleven of twelve months, rising to 33.2% in 2025-05 and 33.5% in 2025-12 [102], coinciding with certified data only in 2025-08 at 33.15% [11]. This divergence reveals that the highest reported efficiencies reflect unverified, sub-0.1 cm² laboratory artifacts where transient electrostatic screening conceals persistent defect vulnerabilities.

A primary mechanistic gap centers on the physical permanence of interfacial contact engineering, where studies diverge on whether molecular modifiers permanently heal deep electronic traps or establish resistive barriers that accelerate degradation under operational bias. At buried hole-transport

layer (HTL) contacts, carbazole phosphonic acid self-assembled monolayer (SAM) molecules optimize dipole alignment and passivate interface states. Those contacts delivered certified 2T tandem efficiencies of 28.78% [3] and 31.47% on 0.945 cm^2 [5]. However, strongly acidic phosphonic acid anchors corrode reactive nickel oxide (NiO_x) contacts [92]. Ultraviolet (UV) illumination irreversibly degrades monolayer hole extraction, with T80 limited to 500 h on an uncertified 26.0% cell [38], and precursor micellar agglomerates of roughly 460 nm cause dewetting pinholes [34,110,138]. At the top heterojunction, wide-bandgap two-dimensional (2D) perovskite capping layers suppress surface non-radiative recombination to reach a certified 26.96% PCE [22] or champion 26% [71]. Yet these low-dimensional overlayers introduce anisotropic out-of-plane transport barriers that trap photo-generated charges, and under operational illumination unshielded internal electric fields lower the activation energy for halide migration, accelerating phase demixing and vacancy aggregation under continuous maximum power point tracking (MPPT).

A parallel gap governs phase stability across tandem absorbers and large-area modules: in 2T tandems, current matching requires mixed-halide wide-bandgap (WBG) absorbers that remain susceptible to photoinduced halide segregation [167,168,169]. In narrow-bandgap (NBG) mixed tin-lead (Sn-Pb) subcells, chemical defect passivation delivers 28.31% certified PCE [6]. However, unrelaxed lattice microstrain drives phase decomposition, necessitating strain-stabilizing networks to achieve certified values of 28.9% [8] and 29.2% [10]. Translating these materials to large modules imposes an uncompromising area penalty, visible across every reported aperture. The largest reported aperture reached 764 cm^2 with a certified PCE of 26.65% [25], and 2T tandems achieved 28.9% certified on 60 cm^2 [18]. Single-junction modules nonetheless dropped to 18.73% certified on 100.15 cm^2 [136] or 24.0% uncertified on 100 cm^2 [137]. None of these large-area certified devices demonstrated accredited operational lifetimes, and conversely the few works reporting T80 metrics rely on disparate thermal and electrical conditions that preclude direct comparison. Those works include 3200 h under ISOS-L-2 for an uncertified 33.5% tandem [102] and 1000 h on a module with certified 25.2% PCE [19]. They also include 1123 h on corrugated substrates with certified 28% PCE [15], and 600 h on a 0.2 cm^2 formamidinium lead triiodide (FAPbI_3) cell [170].

To resolve these contradictions, the field must measure how operational electric fields alter the kinetic activation barrier for ion migration across functional interfaces. This year’s evidence could not support a rigorous comparison between two possibilities: chemical passivators may permanently suppress defect formation, or they may merely provide transient environmental encapsulation that masks accelerated internal field-induced breakdown. Current reporting cannot separate these mechanisms because studies evaluate stability through unencapsulated shelf-storage in dry air or dark nitrogen, while operational tracking also frequently omits temperature regulation, aperture masking, or reverse-bias characterization. Reporting that would have supported this comparison requires measuring the temperature-dependent activation energy of ionic conductivity using operando impedance spectroscopy under calibrated operating bias. That measurement must be paired with continuous MPPT under protocol ISOS-L-2 at $65\text{ }^\circ\text{C}$ or $85\text{ }^\circ\text{C}$ on designated apertures with declared geometric fill factors, alongside pre- and post-aging dark reverse-bias breakdown voltage sweeps. Quantifying ion migration activation energies directly against accredited T80 lifetimes under combined electrical and thermal stress is necessary to determine whether interfacial molecules permanently eliminate recombination centers or simply delay moisture degradation while introducing latent electronic instabilities.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the year, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

AI Usage Declaration

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References

1. Liu, R. et al. Precisely Tuning 3D/Quasi-2D Perovskite Heterojunctions in Wide-Bandgap Perovskites for High-Performance Tandem Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202504321>
2. Jia, X. et al. Defect passivation and crystallization modulation in methylammonium-free wide-bandgap perovskites for all-perovskite tandem solar cells. *Science Advances* (2025). <https://doi.org/10.1126/sciadv.adv4501>
3. Shi, C. et al. Modulating competitive adsorption of hybrid self-assembled molecules for efficient wide-bandgap perovskite solar cells and tandems. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-58111-y>
4. Jin, M. et al. Bidentate Anchoring Enables Concurrent Grain Orientation and Lattice Strain Mitigation in Wide-Bandgap Perovskites for High-Performance All-Perovskite Tandem Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202513281>

5. Guan, Z. et al. Self-Consistent Cation–Anion Integration Eliminates Passivation Crosstalk for Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202509782>
6. Liu, Y. et al. Synergistic immobilization of ions in mixed tin-lead and all-perovskite tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-58810-6>
7. Chen, J. et al. Lewis Base Strategy for Crystallization Control and Buried Interface Passivation on Hydrophobic PTAA Substrate for Efficient Tin–Lead Perovskite and All-Perovskite Tandem Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenerylett.4c03370>
8. Pan, H. et al. Decoupling photothermal-mechanical degradation through lattice-stabilizing networks in Sn–Pb perovskites and all-perovskite tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-64274-5>
9. Liao, X. et al. Methylammonium-free, high-efficiency, and stable all-perovskite tandem solar cells enabled by multifunctional rubidium acetate. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-56549-8>
10. Bai, Y. et al. Lattice stabilization and strain homogenization in Sn-Pb bottom subcells enable stable all-perovskite tandems solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-62661-6>
11. Zhang, D. et al. Iceberg-like pyramids in industrially textured silicon enabled 33% efficient perovskite-silicon tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-62389-3>
12. Wu, L. et al. Crystallization Modulation of Wide-Bandgap Perovskites on Textured Silicon for Tandem Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenerylett.5c01244>
13. Yao, Y. et al. Oriented wide-bandgap perovskites for monolithic silicon-based tandems with over 1000 hours operational stability. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-024-55377-6>
14. Lian, X. et al. A supramolecular approach to improve the performance and operational stability of all-perovskite tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-62391-9>
15. Pei, F. et al. Perovskite/CIGS Tandem Solar Cells with over 1000 h Operational Stability through Interconnection Stress Relief. *Journal of the American Chemical Society* (2025). <https://doi.org/10.1021/jacs.5c13264>
16. Wang, L. et al. Efficient perovskite/Cu(In,Ga)Se₂ tandem solar cells with a composite intermediate recombination layer. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-67350-y>
17. Zhao, H. et al. Synergistic Self-Assembled Monolayers Reinforce Buried Interface Anchoring for High-Efficiency Tandem Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202504237>
18. Artuk, K. et al. 60 cm² perovskite-silicon tandem solar cells with an efficiency of 28.9% by homogeneous passivation. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-63673-y>
19. Urieta-Mora, J. et al. Spiro-Phenothiazine Hole-Transporting Materials: Unlocking Stability and Scalability in Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202505475>
20. Wang, Y. et al. A Soft Nonpolar-Soluble Two-Dimensional Perovskite for General Construction of Mixed-Dimensional Heterojunctions. *Advanced Materials* (2025). <https://doi.org/10.1002/>

adma.202419750

21. Su, Y. et al. Chemical inhibition of light-induced decomposition by hindered amine for efficient and stable perovskite solar cells. *eScience* (2025). <https://doi.org/10.1016/j.esci.2025.100451>
22. Li, G. et al. Stabilizing high-efficiency perovskite solar cells via strategic interfacial contact engineering. *Nature Photonics* (2025). <https://doi.org/10.1038/s41566-025-01791-1>
23. He, D. et al. Universal Ion Migration Suppression Strategy Based on Supramolecular Host–Guest Interaction for High-Performance Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202505115>
24. Yu, F. et al. Homogenized Self-Assembled Molecules for Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202505876>
25. Ding, J. et al. Stabilizing buried interface by bilateral bond strength equilibrium strategy toward efficient perovskite photovoltaics. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-63389-z>
26. Yuan, S. et al. Conjugated Bisphosphonic Acid Self-Assembled Monolayers for Efficient and Stable Inverted Perovskite Solar Cells. *Journal of the American Chemical Society* (2025). <https://doi.org/10.1021/jacs.5c05801>
27. Yang, L. et al. Improving Charge Extraction and Operational Stability in Inverted Perovskite Solar Cells with Conjugated Block Copolymers. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202509065>
28. Yang, J. et al. Flexibility meets rigidity: a self-assembled monolayer materials strategy for perovskite solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-62388-4>
29. Yu, L. et al. Unraveling the Role of Molecular Flexibility in Enhancing Pyridine Based Defect Passivation for Efficient and Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202524079>
30. Yuan, M. et al. Redox-Improved Self-Assembled Monolayers for Inverted Perovskite Solar Cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-66421-4>
31. Wang, Y. et al. Covalent Tridentate Molecule Anchoring Enhances Nickel Oxide for Efficient Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202507730>
32. Lv, Z. et al. Pre-Anchored Ionic Bond Mediators Enabling Controllable Monolayer Assembly for High-Performance Perovskite Solar Cells and Modules. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202513600>
33. Zhang, L. et al. Benzylphosphonic Acid-Engineered Compact Self-Assembled Monolayers for Bifacial Buried Interface Passivation in High-Performance Inverted Perovskite Solar Cells. *Advanced Science* (2025). <https://doi.org/10.1002/advs.202512117>
34. Ma, J. et al. Homogenizing Self-Assembled Monolayers via Covalent Organic Frameworks for Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202519875>
35. Ding, J. et al. Molecular Doping of Self-Assembled Monolayers for Interface Engineering in Inverted Perovskite Solar Cells and Large-area Bifacial Modules: Enhancing Uniformity, Crystallization, and Stability. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202503252>
36. Feng, B. et al. ITO Nanoparticles to Stabilize the Self-Assembly of Hole Transport Layer in Inverted Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202517573>
37. Zhou, S. et al. Enhancing Chemical Bonding at Buried Interface Enables Improved Performance

- of Perovskite Solar Cell. *Small* (2025). <https://doi.org/10.1002/sml.202504930>
38. Liu, T. et al. Efficient perovskite solar modules enabled by a UV-stable and high-conductivity hole transport material. *Science Advances* (2025). <https://doi.org/10.1126/sciadv.adu3493>
 39. Yang, P. et al. Enlarging moment and regulating orientation of buried interfacial dipole for efficient inverted perovskite solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-024-55653-5>
 40. Wang, P. et al. Electrostatically Enhanced Self-Assembled Monolayers Anchoring for Scalable and Stable Inverted Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202514735>
 41. Cheng, M. et al. Tailoring Buried Interface and Minimizing Energy Loss Enable Efficient Narrow and Wide Bandgap Inverted Perovskite Solar Cells by Aluminum Glycinate Based Organometallic Molecule. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202419413>
 42. Li, C. et al. Multifunctional Organic Bridge at Self-Assembled Molecule/Perovskite Interface Enables High-Performance Inverted Perovskite Solar Cells. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202520372>
 43. Li, L. et al. Buried Interface Chelating Molecular Bridge Strategy Enables Highly Efficient and Stable Inverted Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202518406>
 44. Gao, Q. et al. Enhanced interface regulation via π -conjugated heterojunctions for high-efficiency inverted perovskite solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d5ee06342f>
 45. Pan, J. et al. Strong Electron-Withdrawing Molecules Facilitating π -Stacking and Charge Transfer Complexes at Buried Interface and Enabling an Inverted PSC with Open-Circuit Voltage of 1.2 V. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202514365>
 46. Song, F. et al. Buried Interface Modification for High Performance and Stable Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202516012>
 47. Cao, Y. et al. Buried interface modification for high performance and stable perovskite solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d4ee05466k>
 48. Hao, C. et al. Employment of 1-Citrulline as an Effective Molecular Bridge for Regulating the Buried Interface of Perovskite Solar Cells to Achieve High Efficiency and Good Stability. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202508169>
 49. Sun, W. et al. Buried Interface Modification for Reduced Open-Circuit Voltage Loss in Perovskite Solar Cells With Efficiency Exceeding 25.8%. *Carbon Neutralization* (2025). <https://doi.org/10.1002/cnl2.70042>
 50. Wang, D. et al. Tailoring Dual-Site Defect Passivation Molecules to Minimize Buried Interface Energy Loss for Highly Efficient and Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202509529>
 51. Deng, L. et al. Converting Residual Unstable PbI₂ into Robust 2D Perovskite via Spacer Cation Engineering Toward Operationally Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202516433>
 52. Li, Z. et al. Cation- π Synergy Induced Self-Assembly of Semiconductor Spacers for High Efficiency and Stable 2D/3D Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202511235>
 53. Kim, C. et al. Tilted-Oriented 2D Perovskites for High-Performance Solar Cells. *Advanced*

- Energy Materials* (2025). <https://doi.org/10.1002/aenm.202503780>
54. Ahmad, S. et al. Sequential Organic Ligand Modifications to Dedicatedly Restructure Grain Boundary and Surface of Perovskites. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-66478-1>
 55. Wang, R. et al. Stable and Efficient Indoor Photovoltaics Through Novel Dual-Phase 2D Perovskite Heterostructures. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202419573>
 56. Liu, P. et al. Achieving Highly Efficient and Stable Inverted Perovskite Solar Cells through Synergistic Bimolecular Interface Defect Passivation. *ACS Applied Energy Materials* (2025). <https://doi.org/10.1021/acsaem.5c02987>
 57. Li, B. et al. Nanoscale soft interaction-engineered perovskite heterojunctions for highly efficient and reproducible solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-64550-4>
 58. Wang, Y. et al. Bifunctional Ligand-Mediated Dual-Site Passivation Enables High-Performance Perovskite Solar Cells with Efficiency Exceeding 26%. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202510458>
 59. Li, K. et al. Dual-Ligand Regulation of Homogeneous Cation–Anion Distribution Enables Efficient and Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202518592>
 60. Zhao, H. et al. Intermolecular Interaction Induced Synergistic Uniform Passivation of Grain Boundary Multiple Defects Enables High-Performance Inverted Perovskite Solar Cells. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202504424>
 61. Zhang, Q. et al. Multifunctional Butylammonium Acetate Additive Regulates FAPbI 3 Film Growth and Defect Passivation for Highly Efficient and Stable Perovskite Solar Cells with 26.12% Efficiency. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202507934>
 62. Ding, Y. et al. Bulk Passivation of Perovskite Films With Phthalocyanine Derivative for Enhanced Perovskite Solar Cells Efficiency and Stability. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202425443>
 63. Hu, J. et al. Revealing the Critical Role of Electron-Withdrawing Cores in Bulk Passivation of Diammonium Ligands Toward High-Performance Perovskite Solar Cells. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202424259>
 64. Gao, S. et al. Over 26% and 24%-Efficiency Rigid and Flexible Perovskite Photovoltaics through Enhanced Crystallinity and Alleviated Residual Strains. *ACS Nano* (2025). <https://doi.org/10.1021/acsnano.5c09464>
 65. Feng, Z. et al. Multi-Site Lead Passivation via Spatial Configuration Modulation of Additives for Efficient Perovskite Solar Cells. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202502409>
 66. Du, S. et al. Spatially Flexible Multisite Amide Derivative Enhances the Performance of Inverted Perovskite Solar Cells and Modules. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202503781>
 67. Liu, C. et al. Omnidirectional Multi-Type Defect Passivation Enables Efficient and Stable Perovskite Solar Cells via Dual Organic Potassium Salts. *Small* (2025). <https://doi.org/10.1002/sml.202511139>
 68. He, Z. et al. Decoupling Photoinduced Lattice Evolution via Grain Spatial Isolation for Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202511062>

69. Liu, N. et al. Nucleation Engineering to Strengthen Interface Contacts in Single-Crystal Perovskite Photovoltaics. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202500947>
70. You, S. et al. C 60 -based ionic salt electron shuttle for high-performance inverted perovskite solar modules. *Science* (2025). <https://doi.org/10.1126/science.adv4701>
71. He, D. et al. Compacting Molecular Stacking and Inhibiting Self-Aggregation in Fullerene Transporting Layer for Efficient and Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202502950>
72. Li, C. et al. Anchoring ligand engineering enables highly stable MA-free perovskite solar cells with a minimal V OC deficit of 0.32 V. *Energy & Environmental Science* (2025). <https://doi.org/10.34734/fzj-2026-01040>
73. Jiang, H. et al. -Conjugated Dipolar Structures: Synergistic Dipole Superposition for Cathode Modification toward Ohmic Contact and Defect Passivation in Solar Cells. *Journal of the American Chemical Society* (2025). <https://doi.org/10.1021/jacs.5c13145>
74. Wan, Z. et al. Bipyridine–Thiosumanene Isomeric Lewis Bases for Synergistic Defect Passivation and Hole Extraction Enables Over 26% Efficient Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202510255>
75. Liu, H. et al. Energetics Modulation for Efficient and Stable n–i–p Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202502117>
76. Wei, J. et al. Molecular Design-Driven Interface Engineering Enabling Simultaneous Defect Passivation and Enhanced Hole Extraction in Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202513869>
77. Zhang, X. et al. Reducing the Voc Loss of Hole Transport Layer-Free Carbon-Based Perovskite Solar Cells via Dual Interfacial Passivation. *Nano-Micro Letters* (2025). <https://doi.org/10.1007/s40820-025-01775-4>
78. Ahmad, W. et al. Design of high-efficiency quasi-interdigitated back-contact perovskite solar cells via layer optimization and interface engineering. *Chemical Engineering Journal* (2025). <https://doi.org/10.1016/j.cej.2025.171437>
79. Peng, C. et al. SCAPS-1D Simulation of Various Hole Transport Layers’ Impact on CsPbI Br Perovskite Solar Cells Under Indoor Low-Light Conditions. *Preprints.org* (2025). <https://doi.org/10.20944/preprints202505.0921.v1>
80. Mustafa, Z. et al. Theoretical Aspects of Ba AsNbO Based Efficient Photovoltaic Device: Design and Optimization. *Preprint* (2025). <https://doi.org/10.1364/opticaopen.30738665>
81. Yuan, C. et al. Redox Shuttle-Stabilized Tin–Lead Perovskites for All-Perovskite Tandem Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenenergylett.5c01995>
82. Jin, X. et al. All-in-One Molecule Regulated Buried Interface and Crystallization of Tin–Lead Perovskite for Efficient All-Perovskite Tandem Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202518800>
83. Chen, W. et al. Universal in situ oxide-based ABX₃-structured seeds for templating halide perovskite growth in All-perovskite tandems. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-57195-w>
84. Zhang, W. et al. From Bottleneck to Breakthrough: A Materials-Interface Roadmap for Commercial Monolithic All-Perovskite Tandem Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenenergylett.5c03339>
85. Guan, H. et al. Challenges and perspectives toward wide-bandgap perovskite subcell in four-terminal all-perovskite tandem solar cells. *DeCarbon* (2025). <https://doi.org/10.1016/j.decarb>

.2025.100098

86. Lü, X. et al. Organic Dyes for Interfacial Regulation of Iodine in Metal Halide Perovskite Wide-Bandgap and Tandem Solar Cells. *ACS Nano* (2025). <https://doi.org/10.1021/acsnano.5c03542>
87. Zheng, X. et al. Suppressing phase segregation and nonradiative losses by a multifunctional cross-linker for high-performance all-perovskite tandem solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d4ee02898h>
88. Chen, C. et al. Synergistic Dipole-Defect Engineering via Sulfonic Molecular Bridge Boosts Voltage in Wide-Bandgap Perovskite and All-Perovskite Tandem Solar Cells. *ACS Nano* (2025). <https://doi.org/10.1021/acsnano.5c04978>
89. Zhao, Y. et al. Interface parallel dipole regulation in all-perovskite tandem solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d5ee02696b>
90. Jiao, W. et al. Molecular Bridging at Buried Interface Enables Efficient Wide-Bandgap Perovskite Solar Cells. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202501556>
91. Guo, Y. et al. Contact Potential Homogenization via Buried Interface Engineering Enables High-Performance Wide-Bandgap Perovskite Photovoltaics. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202500168>
92. Wang, J. et al. Less-acidic boric acid-functionalized self-assembled monolayer for mitigating NiOx corrosion for efficient all-perovskite tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-59515-6>
93. Fu, S. et al. In situ molecular compensation in wide-bandgap perovskites for efficient all-perovskite tandem solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d5ee01369k>
94. Pu, D. et al. Imprisoning 2H intermediate phases in blade-coated wide-bandgap perovskites for efficient all-perovskite tandem solar cells. *Science Advances* (2025). <https://doi.org/10.1126/sciadv.ady3621>
95. Kumar, P. et al. Recent progress in all-perovskite tandem solar cells and modules: redefining limits. *Progress in Materials Science* (2025). <https://doi.org/10.1016/j.pmatsci.2025.101560>
96. Zhang, Q. et al. Advancing all-perovskite two-terminal tandem solar cells: optimization of wide- and narrow-bandgap perovskites and interconnecting layers. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d4ee06027j>
97. Fitzsimmons, M.R. et al. Optimized Graphene-Oxide-Based Interconnecting Layer in All-Perovskite Tandem Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenrgylett.4c03065>
98. Wei, J. et al. An efficient ALD- and noble metal-free charge recombination architecture for monolithic all-perovskite tandem solar cells. *Journal of Materials Chemistry A* (2025). <https://doi.org/10.1039/d5ta04259c>
99. Wang, J. et al. Exposing binding-favourable facets of perovskites for tandem solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d5ee02462e>
100. Wang, J. et al. Performance and stability analysis of all-perovskite tandem photovoltaics in light-driven electrochemical water splitting. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-024-55654-4>
101. Faiaad, R.M. et al. Unveiling architectural and optoelectronic synergies in lead-free perovskite/perovskite/kesterite triple-junction monolithic tandem solar cells. *Figshare* (2025). <https://doi.org/10.48550/arxiv.2511.06059>

102. Han, Z. et al. Uniform phase distribution of wide bandgap perovskite for high-performance perovskite-silicon tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-66480-7>
103. Zhu, Y. et al. Ionic Site Competition Strategy Enhances Wide-Bandgap Perovskite Stability for Silicon-Based Tandem Photovoltaics. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenenergylett.5c02408>
104. Tang, L. et al. Record-Efficient Flexible Monolithic Perovskite–CIGS Tandem Solar Cell with V OC Exceeding 1.8 V on Polymer Substrate. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202403682>
105. Yang, D. et al. Enhancing the Performance of Perovskite/Silicon Tandem Solar Cells via Cation- Interaction. *Solar RRL* (2025). <https://doi.org/10.1002/solr.202500173>
106. Zhou, X. et al. Knoevenagel Condensation Enables a High-Dipole Interfacial Molecule for Efficient Wide-Bandgap Perovskite and Tandem Photovoltaics. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202518169>
107. Park, S.J. et al. Tailored self-assembled monolayer molecules for perovskite/PERC tandem solar cells with efficiencies over 30%. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d5ee02650d>
108. Chang, G. et al. Bifacially Reinforced Self-Assembled Monolayer Interfaces for Minimized Recombination Loss and Enhanced Stability in Perovskite/Silicon Tandem Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202504520>
109. Wang, W. et al. Super -Hydrophilic SAMs via Chlorine Substituted All-Hydrophilic Molecule Modification for Monolithic Perovskite-Silicon Tandem Solar Cells. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202502222>
110. Liu, L. et al. Micelle-Assisted Formation of Self-Assembled Monolayers for Efficient and Stable Perovskite/Silicon Tandem Solar Cells. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202405675>
111. Zhang, L. et al. Ti3C2Tx/Self-Assembled Monolayer Composite Interface for Enhanced Hole Transport in Inverted Perovskite Solar Cells. *ACS Applied Nano Materials* (2025). <https://doi.org/10.1021/acsanm.5c00290>
112. Raoui, Y. et al. Synergic MXene and S-benzyl-L-cysteine Passivation Strategies for Wide Bandgap Perovskite Solar Cells for 4T Tandem Applications. *Small* (2025). <https://doi.org/10.1002/sml.202411310>
113. Chai, W. et al. Buried Interface Regulation with TbCl3 for Highly-Efficient All-Inorganic Perovskite/Silicon Tandem Solar Cells. *Nano-Micro Letters* (2025). <https://doi.org/10.1007/s40820-025-01763-8>
114. Wang, X. et al. Multi-site passivation agent for efficient tandem solar cells: Simultaneously suppressing defect recombination in NiO surface, perovskite buried interface, and silicon edge. *eScience* (2025). <https://doi.org/10.1016/j.esci.2025.100471>
115. Feng, Z. et al. Buried Interface Modulation Using Self-Assembled Monolayer and Ionic Liquid Hybrids for High-Performance Perovskite and Perovskite/CuInGaSe 2 Tandem Photovoltaics. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202412692>
116. Bì, H. et al. Lead-Free Perovskite Tandem Solar Cells with Wide Bandgap Tin Perovskite and CIGS. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenenergylett.5c00163>
117. Shabanloo, M. et al. Green Dual-Interface Passivation with Succinic Anhydride Derivation for High-Performance Perovskite Solar Cells. *ACS Applied Energy Materials* (2025). <https://doi.org/10.1021/acsaem.5c02901>

118. Tian, J. et al. Overcoming optical losses in thin metal-based recombination layers for efficient n-i-p perovskite-organic tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-024-55376-7>
119. An, Y. et al. Balancing carrier transport in interconnection layer for efficient perovskite/organic tandem solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-58047-3>
120. Sun, X. et al. From 20% single-junction organic photovoltaics to 26% perovskite/organic tandem solar cells: self-assembled hole transport molecules matter. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d4ee05533k>
121. Heydarian, M. et al. Indium-Free Recombination Layer for Perovskite-Based Multijunction-Solar-Cells- with Improved Performance Using Sputtered Zinc Tin Oxide. *Small* (2025). <https://doi.org/10.1002/sml.202511646>
122. Liu, C. et al. A simultaneous synergistic protection mechanism in hybrid perovskite-organic multi-junctions enables long-term stable and efficient tandem solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.34734/fzj-2025-05790>
123. Chen, H. et al. Difunctional Polymerizable Additive Enables Efficient and Stable Wide-Bandgap Perovskites for Perovskite/Organic Tandems Solar Cells. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202502422>
124. Aslam, F. et al. Dual-Interface Passivation of Wide-Bandgap Perovskite Films for Efficient Four-Terminal Perovskite-Organic Tandem Solar Cells. *The Journal of Physical Chemistry Letters* (2025). <https://doi.org/10.1021/acs.jpcllett.5c00756>
125. Alidoust, S. et al. Interface Modeling of Perovskite Polymer Heterostructures for Enhanced Charge Transfer Efficiency in Hybrid Photovoltaic Materials. *arXiv (Cornell University)* (2025). <https://doi.org/10.48550/arxiv.2512.22321>
126. Li, X. et al. Top-Down Dual-Interface Carrier Management for Highly Efficient and Stable Perovskite/Silicon Tandem Solar Cells. *Nano-Micro Letters* (2025). <https://doi.org/10.1007/s40820-024-01631-x>
127. Ji, S. et al. Cooperative Dipole Engineering Unlocks 92.8% Shockley-Queisser Voltage Limit in Wide-Bandgap Perovskites for Tandem Photovoltaics. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenenergylett.5c02738>
128. Zhang, Z. et al. Rationally designed universal passivator for high-performance single-junction and tandem perovskite solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-56068-6>
129. Guo, X. et al. Oblique-Angle Damage-Free Evaporation of Silicon Oxide Electron-Selective Passivation Contacts for Efficient and Stable Perovskite and Perovskite/TOPCon Tandem Solar Cells (Adv. Energy Mater. 4/2025). *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202570020>
130. Jang, Y.H. et al. Sacrificial layer concept interface engineering for robust, lossless monolithic integration of perovskite/Si tandem solar cells yielding high fill factor of 0.813. *Nano Convergence* (2025). <https://doi.org/10.1186/s40580-025-00492-3>
131. Luo, H. et al. Damp-Stable Perovskite/Silicon Tandem Solar Cells with Internal Encapsulating Sulfonium-Based Molecules. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenenergylett.5c01010>
132. Wan, Z. et al. Multifunctional MXene for Thermal Management in Perovskite Solar Cells. *Nano-Micro Letters* (2025). <https://doi.org/10.1007/s40820-025-01855-5>
133. Park, N. Discovery and Progress of Solid-State Perovskite Solar Cells. *Accounts of Chemical*

- Research* (2025). <https://doi.org/10.1021/acs.accounts.5c00612>
134. Raitani, K. et al. Prospects for market-specific design of Perovskite-Silicon tandem solar cells. *arXiv (Cornell University)* (2025). <https://doi.org/10.48550/arxiv.2502.05465>
 135. Tao, Y. et al. Residual Solvent Management Enables High-Efficiency Antisolvent-Free Perovskite Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenrgylett.5c01538>
 136. Jiang, Z. et al. Solvent Engineering for Scalable Fabrication of High-Quality Formamidinium Cesium-Based Perovskite Films Toward Highly Efficient and Stable Solar Modules. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202500598>
 137. Chu, Z. et al. Laser annealing enables rapid, degradation-free ambient processing of perovskite solar modules. *Science* (2025). <https://doi.org/10.1126/science.adx9650>
 138. Zhang, Z. et al. Monodisperse Regulation of Self-Assembled Monolayer Via Dipole Molecules for Efficient Perovskite Solar Cells. *Angewandte Chemie International Edition* (2025). <https://doi.org/10.1002/anie.202512660>
 139. Wei, J. et al. Blade-Coating-Compatible Strategy for Uniform Anchoring of Self-Assembled Molecules toward Scalable Perovskite Solar Module Fabrication. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202502639>
 140. Yin, R. et al. Coassembled Self-Assembled Monolayers Enable Efficient and Stable Inverted Perovskite Solar Cells. *ACS Nano* (2025). <https://doi.org/10.1021/acsnano.5c07208>
 141. Sun, J. et al. Pyridine-assisted solvent engineering for high-quality narrow-bandgap perovskites in efficient tandem modules. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-64038-1>
 142. Xu, B. et al. Reversing the Reaction Order Between FA + and Rb + Enhances the Photovoltaic Performance of Blade-Coated Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202506291>
 143. Han, Y. et al. Synergistic Regulation of Perovskite Crystallization and Defect Passivation via L-Ascorbic Acid Sodium Salt for High Performance FAPbI₃ Perovskite Solar Cells. *The Journal of Physical Chemistry Letters* (2025). <https://doi.org/10.1021/acs.jpcllett.5c01590>
 144. Li, M. et al. In Situ Impurity Phase Repair Strategy Enables Highly-Efficient Perovskite Solar Cells with Periodic Photovoltaic Performance. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202501057>
 145. Chu, L. et al. Methylammonium-Free Perovskite Photovoltaic Modules. *ACS Nano* (2025). <https://doi.org/10.1021/acsnano.4c18089>
 146. Siegrist, S. et al. Unveiling the Role of Cl Incorporation Enables Scalable MA-Free Triple-Halide Wide-Bandgap Perovskites for Slot-Die-Coated Photovoltaic Modules. *Solar RRL* (2025). <https://doi.org/10.1002/solr.202400750>
 147. Chen, M. et al. Reinforced Anchor for Scalable Self-Assembled Monolayer to Attain High-Performance Perovskite Solar Modules. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202502000>
 148. Hu, S. et al. Scalable Impregnation Method for Preparing a Self-Assembled Monolayer in High-Performance Vapor-Deposited Lead-Halide Perovskite Solar Cells. *ACS Nano* (2025). <https://doi.org/10.1021/acsnano.5c01479>
 149. Cai, Z. et al. A Buried Interface Fastening Approach for Efficient and Flexible Perovskite Photovoltaics. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202505921>
 150. Zhu, X. et al. Restrictive Heterointerfacial Delamination in Flexible Perovskite Photovoltaics Using a Bifacial Linker. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202419329>

151. Yang, W.L. et al. Tailoring interface-anchoring molecules for efficient and stable perovskite solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-66225-6>
152. Bai, D. et al. Formamidinium In Situ Assistance for Buried Interfaces in Perovskite Solar Cells. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202501206>
153. Wan, Z. et al. Suppressing Ion Migration through Dual Interface Engineering toward Efficient and Stable Perovskite Solar Modules. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenergylett.5c00074>
154. Sun, Y. et al. In situ coordinated HTL strategy for high-performance and scalable perovskite solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-64111-9>
155. Zhai, M. et al. Fluorene-Terminated π -Conjugated Spiro-Type Hole Transport Materials for Perovskite Solar Cells. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenergylett.4c03233>
156. Zhang, J. et al. Synergistic versatile bistriflimide salts in light-accelerated spiro-OMeTAD oxidation and perovskite module photovoltaics engineering. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-66752-2>
157. Sun, X. et al. Reducing Inter-grain Gaps to Stabilize the Outdoor Operation of Perovskite Solar Modules. *ACS Energy Letters* (2025). <https://doi.org/10.1021/acsenergylett.5c01711>
158. Ding, Y. et al. Synergistic Bulk and Interface Passivation via Conjugated Ionic Additives Enables 26% Efficient PTAA-Based Perovskite Solar Cells. *Advanced Energy Materials* (2025). <https://doi.org/10.1002/aenm.202504647>
159. Chen, X. et al. Entropy-driven strategy stabilizes photoactive halide perovskites for inverted solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-64728-w>
160. Wang, H. et al. 2D Capping Layer Passivation toward Inorganic CsPbI₃ Perovskite Minimodule. *Advanced Functional Materials* (2025). <https://doi.org/10.1002/adfm.202423397>
161. Zai, H. et al. Wafer-scale monolayer MoS₂ film integration for stable, efficient perovskite solar cells. *Science* (2025). <https://doi.org/10.1126/science.ado2351>
162. Li, M. et al. Radical Molecular Network-Buffer Minimizes Photovoltage Loss in FAPbI Perovskite Solar Cells. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202417289>
163. Hansora, D. et al. Scalable and durable module-sized artificial leaf with a solar-to-hydrogen efficiency over 10%. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-59597-2>
164. Azam, M. et al. Tailoring pyridine bridged chalcogen-concave molecules for defects passivation enables efficient and stable perovskite solar cells. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-55815-z>
165. Su, J. et al. Synergistic Dual-Anchor Passivation at Buried Interfaces Enabling High-Efficiency and Stable Perovskite Solar Cells. *The Journal of Physical Chemistry Letters* (2025). <https://doi.org/10.1021/acs.jpclett.5c02124>
166. Liang, Y. et al. Multiple B-site doping suppresses ion migration in halide perovskites. *Science Advances* (2025). <https://doi.org/10.1126/sciadv.ads7054>
167. Datta, K. et al. Local halide heterogeneity drives surface wrinkling in mixed-halide wide-bandgap perovskites. *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-57010-6>
168. Scarperi, A. et al. Probing Photoinduced Halide Segregation in CH₃NH₃Pb(BrxI_{1-x})₃ Perovskites via in Situ Nuclear Quadrupole Resonance Spectroscopy. *Journal of the American Chemical Society* (2025). <https://doi.org/10.1021/jacs.5c10094>
169. Guo, Y. et al. Unveiling the impact of photoinduced halide segregation on performance degra-

- dation in wide-bandgap perovskite solar cells. *Energy & Environmental Science* (2025). <https://doi.org/10.1039/d4ee05604c>
170. Othman, M. et al. Suppression of Stacking Faults for Stable Formamidinium-Rich Perovskite Absorbers. *Advanced Materials* (2025). <https://doi.org/10.1002/adma.202502142>